Sana'a University

Faculty of Computer and Information Technology

Department of Cyber Security

RESEARCH PROPOSAL

A Generalizable Payload-Free Intrusion Detection Framework for Unseen Attack Detection in Encrypted Traffic via Flow-Level Representation Learning

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Abstract

The rapid growth of encrypted network traffic has significantly changed the way modern network security systems detect and analyze cyber threats. Encryption protocols such as TLS, HTTPS, VPN, and QUIC protect user privacy and data confidentiality; however, they also limit the effectiveness of traditional intrusion detection systems that depend on payload inspection, signature matching, or deep packet inspection. As a result, malicious activities can be hidden inside encrypted communication channels, making threat detection more challenging.

Recent machine learning and deep learning approaches have shown promising results in encrypted traffic classification and intrusion detection. However, many existing approaches are evaluated under closed-set assumptions, rely on dataset-specific features, and depend heavily on labeled known attacks. These limitations reduce their ability to detect unseen attack classes in realistic encrypted network environments, where payload inspection is unavailable and attack behaviors continuously evolve.

This proposal aims to develop a generalizable payload-free intrusion detection framework for detecting unseen attacks in encrypted traffic via flow-level representation learning. The proposed framework will rely on flow-level and NetFlow-like metadata, such as packet counts, byte counts, flow duration, packet timing, and direction-based characteristics, without decrypting or inspecting packet payloads. Representation learning techniques will be used to learn meaningful traffic behavior patterns and support the detection of unseen attack classes.

The study will adopt a quantitative algorithm/system evaluation design using benchmark intrusion detection datasets. The proposed framework will be evaluated using open-set or leave-one-attack-out experimental protocols to simulate unseen attack scenarios. Performance will be assessed using standard metrics such as accuracy, precision, recall, F1-score, false positive rate, AUC, and unseen attack detection rate. The expected contribution of this research is a structured and privacy-preserving IDS framework that improves unseen attack detection in encrypted traffic while supporting fair evaluation and reproducibility.

Keywords

Encrypted Network Traffic; Intrusion Detection System; Payload-Free Detection; Flow-Level Features; NetFlow; Representation Learning; Unseen Attack Detection; Open-Set Evaluation; Machine Learning; Cybersecurity

Executive Research Snapshot

Item	Description
Proposed Title	A Generalizable Payload-Free Intrusion Detection Framework for Unseen Attack Detection in Encrypted Traffic via Flow-Level Representation Learning
Research Area	Cybersecurity, Network Security, Intrusion Detection Systems, Encrypted Traffic Analysis
Research Type	Quantitative Computing Research
Study Family	Algorithm / System Evaluation
Core Problem	Existing IDS models for encrypted traffic often rely on closed-set learning, dataset-specific features, and labeled known attacks, limiting their ability to detect unseen attacks without payload inspection.
Main Research Gap	Lack of a generalizable, payload-free, flow-based IDS framework capable of detecting unseen attack classes in encrypted traffic under open-set or leave-one-attack-out evaluation.
Proposed Solution	A payload-free flow-level representation learning framework for detecting unseen attacks in encrypted network traffic.
Input Data	Flow-level / NetFlow-like metadata extracted from encrypted traffic datasets.
Main Techniques	Feature preprocessing, flow-level feature extraction, representation learning, anomaly / novelty detection, open-set evaluation.
Expected Evaluation Protocol	Benchmark evaluation, baseline comparison, leave-one-attack-out, open-set testing.
Expected Metrics	Accuracy, Precision, Recall, F1-score, False Positive Rate, AUC, Unseen Attack Detection Rate.
Expected Contribution	A structured IDS framework that supports privacy-preserving detection of unseen attacks in encrypted traffic without decryption or payload inspection.

Alignment with Supervisor's Chapter Guides

This proposal is designed according to the research methodology structure for quantitative computing studies. The selected topic belongs to the Algorithm / System Evaluation family because

it focuses on designing and evaluating an intrusion detection framework using datasets, experimental protocols, baseline models, and quantitative performance metrics.

Supervisor's Guide Requirement	**Alignment in This Proposal**
Problem + Purpose + Research Questions	Addressed in Chapter 1 through the research problem, aim, objectives, questions, and hypotheses.
Literature Review and Research Gap	Addressed in Chapter 2 through critical review, comparative literature matrix, taxonomy, synthesis, and gap development.
Quantitative Research Design	Addressed in Chapter 3 through algorithm/system evaluation design.
Dataset / Data Source Selection	Addressed in Chapter 3 through dataset selection strategy and dataset description.
Unit of Analysis and Observation	Addressed in Chapter 3 to distinguish between flow records, sessions, and experimental runs.
Experimental Environment	Addressed in Chapter 3 through software, hardware, versions, libraries, and reproducibility settings.
Experimental Protocol	Addressed through baseline comparison, open-set testing, leave-one-attack-out evaluation, and fair comparison policy.
Evaluation Metrics	Addressed through accuracy, precision, recall, F1-score, FPR, AUC, and unseen attack detection rate.
Validity and Reproducibility	Addressed through evidence logging, fair comparison policy, threats to validity, and reproducibility package.
Ethical Considerations	Addressed through payload-free analysis, public datasets, privacy preservation, and no decryption.

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List of Abbreviations

Abbreviation	Meaning
AUC	Area Under the Curve
AUPR	Area Under the Precision-Recall Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
CNN	Convolutional Neural Network
DL	Deep Learning
DPI	Deep Packet Inspection
FNR	False Negative Rate
FPR	False Positive Rate
GPF-UD-ENT	Generalizable Payload-Free Unseen Detection for Encrypted Network Traffic
GRU	Gated Recurrent Unit
IDS	Intrusion Detection System
IPFIX	Internet Protocol Flow Information Export
LOAO	Leave-One-Attack-Out
LSTM	Long Short-Term Memory
ML	Machine Learning
NetFlow	Network Flow Metadata Standard/Format
NIDS	Network Intrusion Detection System
PCAP	Packet Capture
QUIC	Quick UDP Internet Connections
RQs	Research Questions
SOTA	State of the Art
TLS	Transport Layer Security
VPN	Virtual Private Network
VAE	Variational Autoencoder
RF	Random Forest
SVM	Support Vector Machine
KNN	k-Nearest Neighbors
LOF	Local Outlier Factor
OCSVM	One-Class Support Vector Machine
XGBoost	Extreme Gradient Boosting

Chapter 1

Introduction and Research Foundation

1.1 Background and Context

Encrypted network traffic has become a dominant characteristic of modern digital communication. Protocols and technologies such as TLS, HTTPS, VPN tunneling, and QUIC are widely used to protect confidentiality, privacy, authentication, and data integrity during transmission. This protection is essential for legitimate users and organizations; however, it has also changed the visibility available to network security monitoring systems. When packet content is encrypted, traditional inspection techniques cannot directly examine the application payload or search for plaintext attack signatures. The reviewed literature consistently shows that encryption has reduced the effectiveness of classical payload-based monitoring and has motivated a shift toward encrypted traffic analysis based on observable traffic behavior rather than content [1], [2], [3].

Network Intrusion Detection Systems (NIDS) have historically used signature matching, rule-based inspection, protocol fingerprints, and Deep Packet Inspection (DPI) to identify malicious activity. These approaches are effective when the detection logic can inspect payload content or compare traffic to known signatures. In encrypted environments, however, payload inspection is either technically unavailable, computationally costly, legally sensitive, or privacy-invasive. As a result, IDS models that depend on payload content face a structural visibility limitation. Systematic reviews of encrypted traffic anomaly detection emphasize that payload-based evidence used in plaintext traffic cannot be directly applied to encrypted traffic, which makes artificial intelligence, statistical characteristics, and metadata-based analysis increasingly relevant [1], [2].

At the same time, encryption does not remove all observable signals from network communication. Even when payload content is hidden, traffic still exposes non-content characteristics such as flow duration, packet counts, byte counts, packet length statistics, packet direction, tim-

ing behavior, inter-arrival time, burst patterns, and session-level behavior. These characteristics can be represented as flow-level or NetFlow-like metadata. Such metadata is attractive for cybersecurity research because it supports privacy-preserving analysis, can be extracted without decryption, and is more realistic for operational network monitoring than content inspection [5], [6], [9].

Machine learning and deep learning approaches have therefore been explored for encrypted traffic classification, encrypted malicious traffic detection, and intrusion detection. Prior studies have used convolutional neural networks, recurrent models, autoencoders, hybrid architectures, and statistical machine learning classifiers to classify encrypted traffic or detect anomalies [11], [12], [17]. However, high classification accuracy on a benchmark dataset does not necessarily imply that an IDS can detect attack behavior that was not present during training. The problem becomes more difficult when attacks are unseen, when datasets use different feature definitions, or when the detection system is evaluated only under closed-set assumptions [4], [5], [8].

The present study is positioned within this context. It focuses on the design and evaluation of a generalizable payload-free intrusion detection framework for detecting unseen attacks in encrypted network traffic using flow-level representation learning. Rather than attempting to decrypt traffic or inspect packet payloads, the proposed study will use flow-level behavior as the primary source of evidence. The central assumption is that encrypted malicious behavior may still leave measurable behavioral traces in network flows, and that representation learning can help transform these traces into useful latent patterns for known and unseen attack detection [7], [13], [15].

1.2 Research Motivation

The motivation for this study arises from three converging needs in modern cybersecurity research. The first is the need for intrusion detection without decryption. Organizations must detect threats in encrypted communication while preserving user privacy and avoiding the operational burden of decrypting network traffic. Payload-free analysis is therefore not only a technical alternative but also a privacy-preserving direction for security monitoring [1], [2], [3].

The second motivation is the continuous evolution of cyberattacks. Attackers may modify communication behavior, use encrypted channels for command-and-control communication, or exploit protocols that provide confidentiality to hide malicious activity. Detection systems trained only on known attacks may fail when the attack class, traffic pattern, or protocol behavior changes. This limitation is especially important in closed-set supervised learning, where the model is typically trained and tested on the same predefined set of classes. Several studies on encrypted traffic and IDS evaluation indicate that closed-world or dataset-specific evaluation can overestimate model performance and provide limited evidence of real-world robustness [4],

[8], [10].

The third motivation is the need for generalizable and reproducible IDS evaluation. Many IDS studies use feature sets that are specific to a single dataset, which makes cross-dataset comparison difficult. Sarhan and colleagues highlight that the lack of standard feature sets prevents fair evaluation of ML-based NIDS across different network scenarios and makes it difficult to assess whether a model generalizes beyond the dataset on which it was trained [5], [6]. This concern directly motivates the use of standardized flow-level or NetFlow-like metadata in the proposed research.

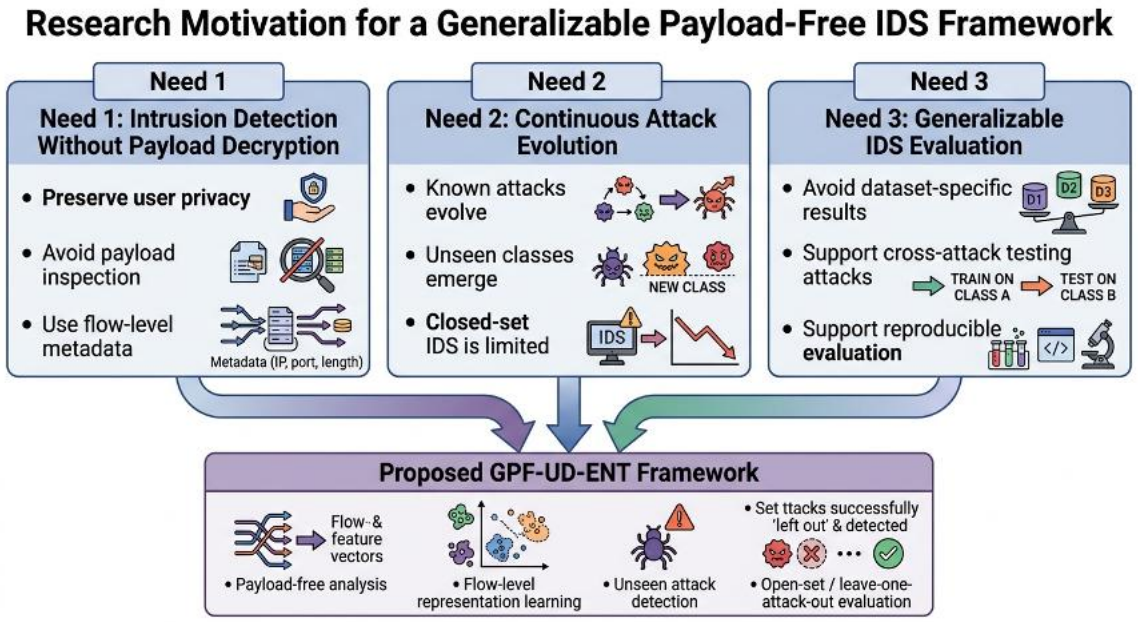


Figure 1.1: Research Motivation for a Generalizable Payload-Free IDS Framework

The study is therefore motivated by the need to connect three research directions that are often treated separately: payload-free encrypted traffic analysis, flow-level feature representation, and unseen attack detection. By integrating these directions into a single algorithm/system evaluation study, the proposed research seeks to develop a framework that can be evaluated under more realistic assumptions than ordinary closed-set classification. The goal is not to claim perfect zero-day detection, but to develop and assess a controlled experimental framework for unseen attack detection using measurable flow-level evidence.

1.3 Research Problem

The research problem addressed in this study is not simply that network traffic is encrypted. Encryption is a legitimate and necessary security mechanism. The deeper problem is that many existing intrusion detection approaches are not sufficiently designed or evaluated for encrypted

environments where payload inspection is unavailable and attack behavior may be unseen during training.

A substantial part of the existing literature reports strong performance for machine learning or deep learning models using benchmark datasets. However, many of these studies are based on closed-set evaluation, where all classes that appear during testing are already represented during training. This setting is useful for known attack classification, but it does not fully reflect realistic environments in which new attack types, modified attack behavior, or unknown encrypted protocols may appear. Studies on encrypted traffic classification and open-world evaluation indicate that models can perform well under closed-world assumptions while still facing challenges when exposed to unseen classes [8], [10].

Another component of the problem is feature dependency. Some IDS models rely on dataset-specific features that are not consistently available across datasets or operational environments. This makes it difficult to compare results fairly and may lead to overfitting to dataset artifacts rather than learning general traffic behavior. The lack of a standard feature set for NIDS datasets has been identified as a major barrier to evaluating generalization across different network scenarios [5], [6].

In addition, anomaly-based and representation learning methods may support unseen attack detection, but they often require careful threshold calibration and may suffer from false positive rates. Excessive false positives reduce operational usefulness and can lead to alert fatigue in security monitoring environments. Recent studies have therefore emphasized temporal behavior modeling, representation learning, and hybrid approaches to improve detection reliability while reducing false alarms [14], [15], [16].

The main research problem can therefore be stated as follows: Existing encrypted traffic intrusion detection approaches remain limited in detecting unseen attacks in a generalizable and payload-free manner because many models depend on closed-set supervised learning, dataset-specific features, known labeled attacks, and evaluation protocols that do not adequately test unseen attack scenarios.

1.4 Problem Decomposition

To clarify the research problem, this study decomposes it into seven connected components. Each component represents a limitation identified in the reviewed literature and has a direct implication for the design of the proposed framework.

Table 1.1: Problem Decomposition Summary

Problem Component	**Description**	**Research Implication**
Encryption Visibility Gap	Encrypted protocols protect packet payloads, preventing traditional IDS techniques from directly inspecting content or searching for plaintext signatures.	The framework must avoid payload inspection and rely on observable traffic behavior instead of decrypted content.
Payload-Free Feature Limitation	When payload features are removed, the available evidence is limited to metadata such as packet sizes, timing, directions, and flow statistics.	The study must identify flow-level and NetFlow-like features that can represent encrypted traffic behavior.
Closed-Set Learning Limitation	Many supervised IDS models assume that all attack classes are known during training and testing.	The evaluation protocol must include unseen attack scenarios rather than only known attack classification.
Unseen Attack Challenge	Attackers may use new, modified, or custom encrypted behaviors that are not represented in training data.	The framework must include a strategy for detecting attack classes excluded from training.
Generalization Problem	Dataset-specific feature sets and traffic artifacts can cause models to perform well on one dataset but poorly elsewhere.	The study must emphasize standardized flow-level features, fair comparison, and generalization-aware evaluation.
Evaluation Weakness	High accuracy under ordinary test splits may not prove robustness against unseen attacks or different traffic conditions.	The methodology must use open-set or leave-one-attack-out evaluation and report metrics beyond accuracy.
Deployment Realism	Operational IDS settings require privacy-preserving analysis, reproducible experiments, and realistic metrics such as false positive rate.	The proposed framework must avoid exaggerated claims and include reproducibility, ethical, and validity considerations.

The decomposition shows that the research problem is multi-dimensional. Encryption creates a visibility gap; payload-free detection limits the available evidence; closed-set learning restricts the ability to identify unseen attacks; and weak evaluation protocols can overstate performance. These components collectively justify the need for a framework that integrates flow-level metadata, representation learning, unseen attack evaluation, and reproducible experimental design.

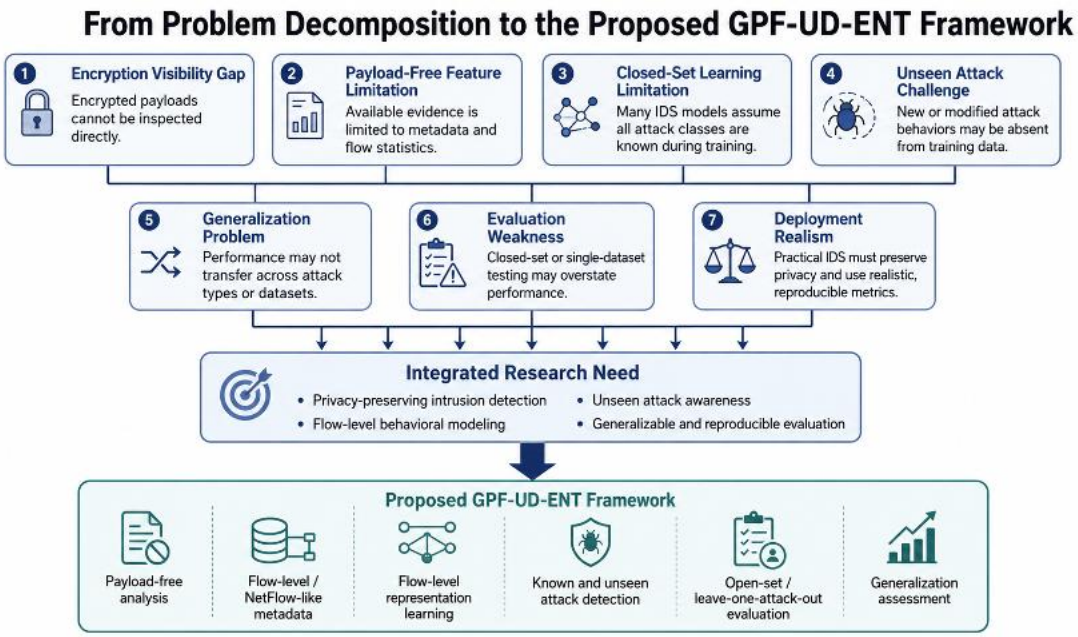


Figure 1.2: Problem Decomposition of Encrypted Traffic IDS Limitations

1.5 Research Gap

The reviewed literature has made important contributions to encrypted traffic analysis and intrusion detection. Surveys and systematic reviews have described the shift from payload inspection to statistical and AI-based encrypted traffic analysis [1], [2], [3]. Experimental studies have demonstrated that deep learning can classify encrypted traffic and identify malicious behavior using packet-level or flow-level patterns [11], [12], [17]. Other studies have emphasized the importance of standard feature sets, NetFlow-based datasets, and fair evaluation across different NIDS datasets [5], [6].

Despite these contributions, the literature still shows an important gap. Many studies focus on known attack detection, application classification, or traffic characterization rather than the detection of attack classes that were excluded from training. Some studies address open-world or unknown class settings, but this line of work remains less developed for payload-free encrypted traffic IDS using standardized flow-level representation learning [7], [8], [10]. Moreover, many evaluations are still closed-set, dataset-specific, or focused on overall accuracy rather than unseen attack detection rate, false positive rate, and generalization evidence [4], [5], [14].

The central research gap is the lack of a generalizable payload-free IDS framework that detects unseen attacks in encrypted network traffic using flow-level representation learning and open-set or leave-one-attack-out evaluation. This gap is methodologically important because it links the feature representation problem, the unseen attack detection problem, and the evaluation problem within one controlled quantitative study.

1.6 Research Aim

This study aims to design and evaluate a generalizable payload-free intrusion detection framework for unseen attack detection in encrypted network traffic using flow-level representation learning.

The aim is intentionally framed around design and evaluation. The study does not claim to solve all zero-day detection problems or to replace operational IDS platforms. Instead, it proposes a research framework that can be experimentally evaluated using benchmark datasets, flow-level metadata, baseline comparison, and open-set or leave-one-attack-out testing.

1.7 Research Objectives

To achieve the research aim, the study will pursue the following objectives:

O1. To analyze the limitations of existing intrusion detection approaches for encrypted network traffic, with emphasis on payload inspection, closed-set learning, dataset-specific features, and unseen attack detection.

O2. To identify suitable payload-free flow-level and NetFlow-like features that can represent encrypted traffic behavior without decrypting packet content.

O3. To design a flow-level representation learning component capable of learning behavioral patterns from encrypted traffic metadata.

O4. To develop a generalizable payload-free intrusion detection framework that integrates flow-level preprocessing, representation learning, known attack detection, and unseen attack detection.

O5. To define an unseen attack detection strategy using open-set or leave-one-attack-out evaluation protocols.

O6. To select benchmark datasets and construct an experimental protocol suitable for evaluating known and unseen attack detection performance.

O7. To compare the proposed framework with selected baseline and state-of-the-art machine learning and deep learning approaches under fair comparison conditions.

O8. To assess the generalization, validity, reproducibility, and limitations of the proposed framework using appropriate metrics and experimental evidence.

1.8 Research Questions

This study is guided by the following research questions:

RQ1: What are the main limitations of existing payload-free intrusion detection approaches for encrypted network traffic?

RQ2: Which flow-level or NetFlow-like features are suitable for representing encrypted network traffic without payload inspection?

RQ3: How can representation learning improve the modeling of encrypted traffic behavior using flow-level metadata?

RQ4: How effective is the proposed framework in detecting unseen attack classes in encrypted network traffic?

RQ5: To what extent does the proposed framework generalize across attack classes, datasets, or traffic conditions compared with baseline models?

RQ6: What limitations, validity threats, and deployment considerations affect the proposed framework?

Table 1.2: Research Questions and Corresponding Objectives

Research Question	**Objective(s)**	**Expected Evidence**
RQ1: What are the main limitations of existing payload-free intrusion detection approaches for encrypted network traffic?	O1	Critical synthesis of the literature, problem decomposition, and limitation analysis.
RQ2: Which flow-level or NetFlow-like features are suitable for representing encrypted network traffic without payload inspection?	O2	Feature selection rationale, dataset feature review, and mapping of available flow-level metadata.
RQ3: How can representation learning improve the modeling of encrypted traffic behavior using flow-level metadata?	O3, O4	Representation learning design, latent feature analysis, and comparison with non-representation baselines.
RQ4: How effective is the proposed framework in detecting unseen attack classes in encrypted network traffic?	O3, O5, O6, O7	Open-set or leave-one-attack-out results using unseen attack detection rate, recall, F1-score, and FPR.

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Research Question	Objective(s)	Expected Evidence
RQ5: To what extent does the proposed framework generalize across attack classes, datasets, or traffic conditions compared with baseline models?	O7, O8	Baseline comparison, generalization experiments, and performance variation across evaluation settings.
RQ6: What limitations, validity threats, and deployment considerations affect the proposed framework?	O8	Threats to validity analysis, reproducibility evidence, and discussion of dataset and deployment constraints.

1.9 Research Hypotheses

Because this study follows a quantitative algorithm/system evaluation design, the following hypotheses are proposed for empirical testing:

H1: Payload-free flow-level and NetFlow-like features contain sufficient behavioral signals to support encrypted traffic intrusion detection without payload inspection.

H2: Flow-level representation learning improves the separability of benign, known attack, and unseen attack behavior compared with conventional feature-based classification alone.

H3: The proposed framework achieves better unseen attack detection performance than selected closed-set supervised baseline models under open-set or leave-one-attack-out evaluation.

H4: Leave-one-attack-out evaluation will reveal performance degradation not captured by closed-set accuracy alone.

These hypotheses are framed as testable expectations rather than guaranteed outcomes. They will be evaluated using benchmark datasets, controlled experimental protocols, baseline comparisons, and quantitative performance metrics.

1.10 Expected Contribution and Novelty

The expected contribution of this study is a structured and experimentally evaluated framework for payload-free unseen attack detection in encrypted network traffic. The proposed novelty lies not in claiming a completely new category of IDS, but in integrating several important components into a coherent and reproducible research framework.

First, the study is expected to provide a generalizable payload-free IDS framework that avoids payload decryption and uses flow-level metadata as the primary input. This supports privacy-preserving analysis and aligns with the limitations of encrypted traffic monitoring [1], [2], [6].

Second, the study is expected to include a flow-level representation learning component. This component is intended to learn compact behavioral representations from traffic metadata, supporting both known attack recognition and unseen attack detection. Prior work on autoencoder-based, deep learning-based, and self-supervised approaches supports the relevance of representation learning for anomaly and encrypted traffic detection [7], [13], [15].

Third, the study is expected to develop an unseen attack detection strategy based on open-set or leave-one-attack-out evaluation. This is important because real-world encrypted traffic may include attack classes not available during training, and closed-set testing alone does not adequately evaluate this condition [8], [10].

Fourth, the study is expected to provide a fair baseline and SOTA comparison plan. The framework will be compared with selected machine learning and deep learning baselines under consistent data splits, feature conditions, and evaluation metrics. This addresses concerns about dataset-specific evaluation and unfair comparison [5], [6].

Finally, the study is expected to deliver a reproducible experimental pipeline, including dataset documentation, preprocessing rules, feature extraction procedures, model settings, random seeds, evaluation protocols, and validity considerations. This contribution supports methodological transparency and supervisor review readiness.

1.11 Significance of the Study

The significance of this study can be described from five perspectives.

Scientific significance. The study contributes to the scientific understanding of encrypted traffic intrusion detection by focusing on unseen attack detection rather than only known attack classification. It connects encrypted traffic analysis, flow-level metadata, representation learning, and open-set evaluation within one research framework.

Practical significance. The proposed framework may support privacy-preserving intrusion detection because it avoids decryption and payload inspection. This is important for organizations that need to monitor encrypted traffic while respecting privacy, compliance, and operational constraints.

Methodological significance. The study emphasizes fair comparison, standardized flow-level features, benchmark datasets, open-set evaluation, and reproducibility. These methodological elements address several limitations identified in ML-based NIDS research, particularly dataset-specific evaluation and weak generalization evidence [5], [6].

Privacy significance. Payload-free detection reduces the need to inspect sensitive content. Although metadata analysis does not eliminate all privacy concerns, it is less intrusive than decrypting and inspecting packet payloads. This makes the proposed direction more compatible with privacy-preserving security monitoring.

Security significance. Detecting unseen attacks in encrypted traffic is a critical challenge because attackers may exploit encrypted channels to hide malicious behavior. A framework that can identify suspicious flow-level behavior without relying only on known labels may improve the resilience of IDS research against evolving threats.

1.12 Scope and Delimitations

The scope of this study is deliberately limited to make the research feasible, measurable, and defensible within a master-level proposal. The following table clarifies what is included and excluded.

Table 1.3: Scope and Delimitations

Included in the Study	**Excluded from the Study**
Encrypted network traffic and encrypted traffic intrusion detection.	Breaking, weakening, or bypassing TLS, VPN, QUIC, HTTPS, or any encryption protocol.
Payload-free detection using flow-level and NetFlow-like metadata.	Payload decryption or deep packet inspection of encrypted content.
Flow/session-level analysis using metadata such as duration, packet counts, byte counts, timing, and direction.	Application-layer content analysis, malware reverse engineering, or host-based forensic analysis.
Representation learning for traffic behavior modeling.	Human-centered surveys, interviews, or user-behavior questionnaires.
Known and unseen attack detection under controlled experimental settings.	Claims of perfect zero-day detection or complete coverage of all future attacks.
Open-set or leave-one-attack-out evaluation using benchmark datasets.	Full production SOC deployment or real-time hardware appliance implementation.
Baseline and SOTA comparison under fair experimental conditions.	Unauthorized traffic collection, active exploitation, or offensive security testing.

These delimitations prevent the study from making exaggerated claims. The research focuses on designing and evaluating a framework under controlled experimental conditions, not on deploying a complete commercial IDS or solving every encrypted traffic detection challenge.

1.13 Limitations

Several limitations are expected in this study. First, the research will rely on publicly available benchmark datasets. Although such datasets enable reproducible evaluation and comparison, they may not fully represent the diversity, scale, and noise of real enterprise networks.

Second, the availability and quality of encrypted traffic labels may affect the evaluation. Some datasets may contain limited encrypted attack categories, incomplete protocol diversity, or labels generated under controlled laboratory conditions.

Third, unseen attack detection will be simulated experimentally, most likely through open-set or leave-one-attack-out evaluation. This strategy is appropriate for controlled research, but it is not identical to real-world zero-day attacks, where the attacker may deliberately adapt behavior to avoid detection.

Fourth, class imbalance may influence results. Rare attack categories may have fewer samples, which can affect model learning, threshold calibration, and evaluation metrics. The study must therefore report more than overall accuracy and include class-sensitive metrics such as recall, F1-score, false positive rate, and unseen attack detection rate [14], [16].

Fifth, feature extraction tools and dataset preprocessing pipelines may produce different flow-level values. Differences in flow timeout settings, aggregation windows, feature definitions, and NetFlow exporters can influence results. This limitation will be addressed through clear documentation of preprocessing and feature extraction procedures.

Sixth, cross-dataset generalization may remain challenging because datasets differ in network topology, traffic generation methods, attack scenarios, and feature distributions. The study will treat generalization as an evaluation objective rather than a guaranteed property.

Seventh, threshold calibration may affect unseen attack detection performance and false positive rates, especially when validation data does not fully represent deployment conditions [Reference needed].

Finally, real-time deployment will not be fully implemented. The study may discuss deployment feasibility, but the main contribution will remain a research prototype and experimental evaluation pipeline.

1.14 Operational Definition of Key Terms

For clarity and consistency, the following terms are operationally defined as they are used in this study.

Table 1.4: Operational Definition of Key Terms

Term	Operational Definition in This Study
Encrypted Network Traffic	Network communication protected by protocols or technologies such as TLS, HTTPS, VPN, or QUIC, where packet payload content is not directly available for inspection.
Intrusion Detection System	A security mechanism that monitors network behavior and attempts to identify benign, malicious, anomalous, or suspicious traffic.
Payload-Free Detection	A detection approach that does not decrypt or inspect packet payload content and instead uses metadata or flow-level characteristics.
Network Flow	A sequence of packets sharing communication attributes such as source, destination, ports, protocol, and temporal boundaries.
Flow-Level Features	Numerical or categorical characteristics extracted from a network flow, such as duration, packet counts, byte counts, packet size statistics, timing, and direction.
NetFlow-like Metadata	Operationally extractable flow metadata similar to NetFlow or IPFIX-style records, used to describe traffic behavior without payload content.
Representation Learning	A machine learning approach that transforms input features into latent or learned representations that capture useful traffic behavior patterns.
Unseen Attack	An attack class or behavior excluded from model training but included during testing to evaluate detection of previously unobserved attack behavior.
Open-Set Detection	An evaluation condition where the model may encounter classes during testing that were not included during training.
Leave-One-Attack-Out Evaluation	An experimental protocol in which one attack class is removed from training and used during testing as an unseen class; the process may be repeated across attack classes.
Generalization	The ability of a model or framework to maintain reliable performance across different attack classes, datasets, traffic conditions, or evaluation splits.
Baseline	A reference model or method used for comparison with the proposed framework, such as Random Forest, SVM, XGBoost, CNN-LSTM, or standalone Autoencoder.
SOTA	State-of-the-art methods selected from the literature for comparison when they are relevant to the same task, dataset type, or evaluation setting.
False Positive Rate	The proportion of benign traffic incorrectly classified as malicious or anomalous.
Unseen Attack Detection Rate	The proportion of samples from excluded attack classes correctly identified as unseen, anomalous, or malicious under the selected evaluation protocol.

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Term	Operational Definition in This Study
Threshold Calibration	The process of selecting decision thresholds using validation data.
Latent Representation	A learned compact feature representation produced by a representation learning model.

1.15 Organization of the Proposal

The proposal is organized into three main chapters. Chapter 1 presents the introduction and research foundation. It establishes the background, motivation, research problem, problem decomposition, research gap, aim, objectives, research questions, hypotheses, expected contribution, significance, scope, limitations, and operational definitions of key terms.

Chapter 2 presents the literature review and research gap development. It reviews encrypted network traffic and security monitoring challenges, network intrusion detection systems, encrypted traffic analysis, payload-based versus payload-free detection, flow-level feature engineering, machine learning and deep learning for encrypted traffic IDS, flow-level representation learning, known versus unseen attack detection, open-set and anomaly detection, generalization, datasets, benchmarking, evaluation metrics, comparative literature matrix, taxonomy of existing approaches, critical synthesis, research gap development, and the conceptual framework of the proposed study.

Chapter 3 describes the research methodology and experimental design. It includes methodological positioning, research design, study family selection, the proposed GPF-UD-ENT framework architecture, dataset selection strategy, data sources, unit of analysis and observation, preprocessing plan, flow-level feature extraction, representation learning design, known and unseen attack detection strategies, detection decision strategy, generalization strategy, baseline and SOTA selection, fair comparison policy, experimental protocol, unseen attack evaluation protocol, variables, metrics, statistical analysis plan, experimental environment, reproducibility package, threats to validity, ethical considerations, feasibility, expected outcomes, and proposed timeline.

Chapter 2

Literature Review and Research Gap Development

2.1 Introduction

This chapter reviews and synthesizes the literature related to the proposed study entitled A Generalizable Payload-Free Intrusion Detection Framework for Unseen Attack Detection in Encrypted Traffic via Flow-Level Representation Learning. The purpose of this chapter is not to provide a descriptive list of previous studies, but to develop a critical and thematic foundation for the research problem, research gap, aim, objectives, questions, hypotheses, and expected contribution established in Chapter 1. Accordingly, the reviewed literature is organized from the broader context of encrypted network traffic and intrusion detection toward the more specific issues of payload-free flow-level representation learning, unseen attack detection, open-set evaluation, and generalization.

The chapter begins by reviewing security monitoring challenges introduced by encrypted traffic, followed by the evolution of network intrusion detection systems (NIDS). It then discusses encrypted traffic analysis without decryption, the distinction between payload-based and payload-free detection, and the role of flow-level and NetFlow-like features. The review then examines machine learning (ML), deep learning (DL), and representation learning approaches for encrypted traffic intrusion detection, with particular attention to the limitations of closed-set evaluation and the challenge of detecting attack classes that are not observed during training. The final sections synthesize the literature into a comparative matrix, taxonomy, research gap development matrix, and conceptual framework that position the proposed study within the current body of knowledge.

The structure of this chapter follows the supervisor's guidance that a literature review should move beyond article-by-article summaries and instead build a coherent synthesis of concepts,

empirical evidence, limitations, and unresolved research gaps. Therefore, each section is linked to the research direction of Chapter 1: payload-free detection, flow-level metadata, representation learning, unseen attack detection, open-set or leave-one-attack-out evaluation, fair baseline comparison, and reproducibility.

2.2 Literature Review Strategy

This study adopts a structured narrative literature review supported by a comparative literature matrix. The review is not presented as a full systematic literature review, but it applies systematic screening logic to ensure that the selected studies are relevant, recent, and aligned with the research problem. The review focuses on peer-reviewed and academically credible studies in encrypted traffic analysis, NIDS, ML/DL-based intrusion detection, flow-level features, representation learning, open-set or novelty detection, and IDS generalization.

The main databases considered for the literature review were IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, and Google Scholar. The primary search period was 2020–2026, while foundational studies published before 2020 were included when they provided essential background, such as early surveys of encrypted traffic classification, deep learning for encrypted traffic, and well-known datasets or feature engineering studies [3], [17], [18].

The literature matrix developed earlier in this proposal was used as the main analytical instrument. Each selected study was examined according to its research problem, traffic type, detection approach, feature type, dataset, evaluation protocol, metrics, findings, limitations, and relevance to the proposed study. This matrix supports the transition from literature review to research gap development.

The comparative matrix focuses on 23 core studies that directly support the research gap, while the reference list also includes supporting methodological, dataset, and baseline references used to justify the experimental design.

Table 2.1: Literature Review Search Strategy

Element	**Description**	**Purpose in This Study**
Databases	IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, and Google Scholar.	To identify peer-reviewed studies in encrypted traffic analysis, IDS, ML/DL, and representation learning.
Search period	Mainly 2020–2026, with selected foundational studies before 2020.	To balance recent research trends with established concepts and baseline references.

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Element	Description	Purpose in This Study
Core keywords	encrypted traffic intrusion detection; payload-free intrusion detection; flow-based IDS; NetFlow intrusion detection; encrypted traffic analysis.	To retrieve studies related to encrypted traffic and payload-free detection.
Advanced keywords	unseen attack detection; zero-day intrusion detection; open-set intrusion detection; novelty detection IDS; representation learning IDS; generalization in intrusion detection.	To identify literature related to the specific research gap of unseen attack detection.
Inclusion criteria	Studies addressing IDS, encrypted traffic, flow-level features, ML/DL, representation learning, open-set/novelty/anomaly detection, benchmarking, or generalization.	To ensure direct relevance to the proposed framework and research questions.
Exclusion criteria	Studies unrelated to network traffic, purely human-centered cybersecurity studies, non-technical opinion papers, and studies with unclear methodological relevance.	To avoid diluting the literature review with sources outside the scope of the proposal.
Screening logic	Studies were screened by title, abstract, methodology, dataset relevance, feature type, evaluation protocol, and stated limitations.	To prioritize sources that can support problem decomposition, gap development, and methodology design.
Literature matrix role	The matrix compares studies by problem, method, data, features, evaluation, limitations, and relevance.	To support thematic synthesis rather than isolated article summaries.
Final confirmed sources	The final review included 23 confirmed sources selected after applying the inclusion and exclusion criteria.	To clarify the scope of the literature review and strengthen the transparency of the review strategy.

2.3 Encrypted Network Traffic and Security Monitoring Challenges

Encrypted network traffic refers to network communication in which packet payload content is protected using encryption protocols such as TLS, HTTPS, VPN, and QUIC. These protocols are essential for protecting confidentiality, privacy, authentication, and integrity during communication. However, the same protection that prevents unauthorized disclosure of content also reduces the visibility available to network monitoring systems and intrusion detection tools.

The reviewed literature consistently indicates that the increasing adoption of encrypted traffic changes the assumptions of traditional security monitoring. Techniques based on deep packet

inspection (DPI), plaintext payload analysis, and content signatures become less effective because the discriminative content is no longer visible to the defender [1], [2], [3]. This creates a monitoring paradox: encryption is necessary for privacy and security, but it also creates a channel in which malicious traffic may hide from inspection-based defenses.

A recent protocol-specific source on TLS 1.3 and QUIC monitoring challenges should be added to strengthen the protocol-level justification for encrypted traffic IDS [Reference needed].

Encrypted traffic analysis therefore requires privacy-preserving alternatives that do not depend on decrypting packet payloads. Surveys and empirical studies show that useful traffic signals can still be extracted from metadata, flow statistics, packet size distributions, timing behavior, directionality, and session-level patterns [1], [2], [3], [4]. This motivates the payload-free focus of the proposed study: the detection process should use observable flow-level behavior rather than encrypted content.

2.4 Network Intrusion Detection Systems

A network intrusion detection system monitors network activity to identify malicious or suspicious behavior. Traditional IDS approaches are often classified into signature-based, anomaly-based, and hybrid detection. Signature-based IDSs compare traffic against known attack patterns and are usually effective for previously observed threats, but they are limited when attacks are modified or unknown. Anomaly-based IDSs model normal behavior and flag deviations, making them more relevant to unseen attack detection, but they can suffer from high false positive rates [16]. Hybrid IDS approaches combine elements of both strategies to improve detection flexibility.

The development of ML- and DL-based IDSs has introduced new capabilities for learning complex network behavior from data. Systematic reviews of DL-based IDS show that neural approaches can improve detection performance, reduce reliance on manual rules, and capture nonlinear traffic relationships [16]. However, many IDS studies still use supervised closed-set classification, where all attack classes appearing during testing are also represented during training. This assumption does not fully match operational environments where new or unseen attack behaviors may emerge.

In encrypted environments, the limitations of traditional IDS become more serious because payload visibility is unavailable. Therefore, IDS design must shift from content inspection toward behavioral analysis. This shift aligns with the proposed study, which treats encrypted traffic intrusion detection as a quantitative algorithm/system evaluation problem based on flow-level metadata and representation learning.

2.5 Encrypted Traffic Analysis

Encrypted traffic analysis aims to infer meaningful information from encrypted communication without decrypting packet content. In network security, this includes identifying application types, detecting abnormal behavior, distinguishing benign and malicious traffic, and supporting intrusion detection under privacy constraints. Prior work has shown that encrypted traffic classification can use statistical and behavioral features such as packet lengths, inter-arrival times, flow duration, packet direction sequences, and header-derived metadata [2], [3], [8], [17], [18].

It is important to distinguish between encrypted traffic classification and encrypted traffic intrusion detection. Traffic classification often focuses on identifying applications, services, protocols, or traffic categories, whereas intrusion detection focuses on identifying malicious or suspicious behavior. Studies such as Deep Packet and Deep-Full-Range demonstrate the relevance of deep learning for encrypted traffic classification and related security tasks [11], [12]. However, classification accuracy in known traffic categories does not automatically prove the ability to detect unseen attacks.

The reviewed literature suggests that encrypted traffic analysis provides a foundation for payload-free IDS, but it also has limitations. Behavioral features may be dataset-dependent, traffic collection procedures may introduce artifacts, and some features may not generalize across protocols, networks, or attack conditions [4], [5], [6]. These limitations justify the need to connect encrypted traffic analysis with standardized flow-level features and generalization-aware evaluation.

2.6 Payload-Based vs Payload-Free Detection

Payload-based detection relies on inspecting packet content, signatures, strings, or application-layer semantics. This approach can be effective when payloads are visible, but it is unsuitable for encrypted traffic unless decryption is available and legally, operationally, and ethically acceptable. In contrast, payload-free detection avoids payload inspection and uses observable metadata such as headers, flows, packet sizes, timing, and statistical behavior. This distinction is central to the proposed study because the research deliberately excludes decryption and DPI of encrypted content.

Table 2.2: Payload-Based vs Payload-Free Intrusion Detection

Detection Approach	Data Used	Strengths	Limitations	Suitability for Encrypted Traffic
Payload-based detection	Packet payload, content signatures, application data.	Can use rich semantic evidence when plaintext is available.	Limited or unusable when payload is encrypted; raises privacy and operational concerns.	Low suitability unless decryption is allowed.
Header-based detection	Protocol headers, ports, flags, limited packet metadata.	Lightweight and easy to extract.	May be insufficient alone; ports and headers can be manipulated or obscured.	Moderate suitability as part of a larger payload-free approach.
Flow-based detection	Flow duration, packet counts, byte counts, timing, packet length statistics, direction ratios.	Practical, scalable, and compatible with encrypted traffic.	Loss of payload semantics; features may depend on dataset and extraction tools.	High suitability for this study.
Payload-free detection	Metadata, flow-level statistics, NetFlow-like records, behavioral features.	Privacy-preserving and avoids decryption; supports encrypted traffic monitoring.	May require stronger representation learning and careful evaluation to detect unseen attacks.	Highest suitability for the proposed framework.

The comparison shows that payload-free flow-based detection is the most appropriate direction for this proposal. It aligns with the constraints of encrypted traffic, supports privacy-preserving analysis, and enables experimental evaluation using flow-level or NetFlow-like datasets [5], [6].

2.7 Flow-Level and NetFlow-Based Feature Engineering

Flow-level features describe communication behavior at the level of a network flow or session rather than individual payload content. Examples include flow duration, packet counts, byte counts, flow rate, inter-arrival time, packet length statistics, forward/backward packet ratios, direction-based features, and burst behavior. NetFlow-like metadata refers to standardized flow records that summarize communication between endpoints using features typically derived from headers and traffic behavior rather than packet content.

Flow records and IPFIX-style summaries are also relevant because they describe communication behavior without requiring access to encrypted payload content, which makes them suitable for payload-free security monitoring [Reference needed].

Flow-level and NetFlow-based features are particularly relevant to encrypted traffic because they can be extracted without breaking encryption. Sarhan et al. proposed standard NetFlow-based feature sets and converted multiple benchmark NIDS datasets into NetFlow variants to support fairer comparison and more practical evaluation [5], [6]. This work is important be-

cause dataset-specific feature sets make it difficult to determine whether a model is genuinely generalizable or merely tuned to a specific dataset representation.

Despite their practical advantages, flow-level features are not a complete solution. They do not capture payload semantics, may be affected by traffic collection tools, and can reflect dataset artifacts. These limitations motivate the use of representation learning, which can transform raw flow-level features into latent behavioral representations that may better support known and unseen attack detection.

2.8 ML for Encrypted Traffic IDS

Machine learning has been widely used for intrusion detection because many NIDS datasets are structured as tabular records with numerical and categorical features. Classical ML models such as Support Vector Machines, Random Forests, k-Nearest Neighbors, Logistic Regression, Decision Trees, and XGBoost are commonly used because they can handle flow-level features and provide strong baseline performance [4], [14], [15], [35], [36].

In encrypted traffic intrusion detection, ML models are attractive because they can operate on statistical and behavioral features without inspecting payload content. Comparative work on encrypted malicious traffic detection shows that ML methods can distinguish benign and malicious encrypted traffic under selected feature sets and datasets [4]. Feature analysis studies also indicate that ML models can help identify which encrypted traffic features are informative, reducing reliance on manual expert intuition alone [9].

These models are treated as baseline candidates in this proposal; final selection will depend on dataset compatibility, feature availability, and experimental feasibility.

However, a recurring limitation is that supervised ML models are often trained and evaluated under closed-set assumptions. They may perform well on known attack classes but remain limited when facing unseen or evolving attacks. They may also overfit to dataset-specific traffic patterns, split policies, or feature extraction artifacts [4], [5], [6]. Therefore, ML models are useful baselines for the proposed study, but they are not sufficient alone to address the research gap.

2.9 Deep Learning for Encrypted Traffic Intrusion Detection

Deep learning has been used in encrypted traffic and IDS research to learn nonlinear and hierarchical patterns from traffic data. Common architectures include multilayer perceptrons, convolutional neural networks, recurrent neural networks such as LSTM and GRU, autoencoders, and transformer-based models. Deep Packet and Deep-Full-Range illustrate how deep learn-

ing can support encrypted traffic classification and intrusion detection-related tasks by learning representations from packet or traffic data [11], [12].

Hybrid DL approaches have also been proposed for IDS. CNN-LSTM models combine spatial and temporal learning, while autoencoder-based approaches learn compact latent representations for anomaly detection [14], [15], [21], [22], [26]. BERT- and CNN-based encrypted traffic classification demonstrates that transformer-style representation learning can capture global traffic features while CNN components capture local patterns [19]. These studies support the broader idea that complex traffic behavior may require learned representations rather than simple handcrafted features.

However, in the context of this study, deep learning is not discussed as a general-purpose classification solution. Its relevance is specifically connected to flow-level representation learning for payload-free encrypted traffic analysis. Since encrypted payloads are unavailable, the model must rely on observable traffic behavior such as flow duration, packet counts, byte counts, packet timing, direction-based features, and NetFlow-like metadata. Deep learning-based representation methods can transform these raw or engineered flow-level features into latent representations that may better capture behavioral differences between benign traffic, known attacks, and potentially unseen attack classes.

The main limitations of DL approaches include the need for large and representative datasets, risk of overfitting, limited explainability, computational cost, and insufficient evaluation under open-set or cross-dataset conditions [16]. Therefore, the proposed study does not treat DL as a guaranteed solution. Instead, it positions flow-level representation learning as a design component that must be evaluated against baselines, unseen attack protocols, and generalization criteria. This establishes the transition to the next section, which focuses specifically on flow-level representation learning as the deep learning direction most aligned with the proposed payload-free IDS framework.

2.10 Flow-Level Representation Learning

Representation learning refers to learning useful internal representations of data that improve downstream tasks such as classification, anomaly detection, or novelty detection. In the context of this study, flow-level representation learning means transforming flow-level or NetFlow-like metadata into latent behavioral representations that may separate benign traffic, known attacks, and unseen attack behavior more effectively than raw features alone.

Autoencoders are one of the most common representation learning approaches in IDS because they can learn compressed representations and reconstruction patterns. Hybrid autoencoder frameworks have been used to combine unsupervised representation learning with supervised classification or probabilistic modeling [15]. Temporal representation methods such as GRU-

VAE and CNN-LSTM can capture sequence or time-dependent characteristics of network traffic, which is important because attacks may appear as changes in temporal behavior rather than isolated feature values [14], [26].

Recent encrypted traffic research also points toward contrastive and self-supervised representation learning. Bai et al. proposed contrastive learning pre-training for encrypted traffic threat detection to reduce dependence on labeled data and improve adaptation to unknown custom encryption protocols [7]. Pidgorniy and Lavryk compared autoencoder, CNN-LSTM, and self-supervised approaches for encrypted traffic anomaly detection, showing the relevance of representation learning to encrypted monitoring without decryption [13]. These works support the proposed study's focus on flow-level representation learning for unseen attack detection.

Nevertheless, representation learning introduces methodological challenges. Latent spaces require careful interpretation, threshold calibration can affect false positives, and improved representation does not automatically imply generalization. Therefore, Chapter 3 will need to specify how representations are learned, how thresholds or decision strategies are selected, and how unseen attack performance is evaluated.

2.11 Known Attack Detection vs Unseen Attack Detection

A central distinction in this proposal is the difference between known attack detection and unseen attack detection. Known attack detection typically assumes a closed-set classification setting in which the attack classes observed during testing are already represented during training. In contrast, unseen attack detection evaluates whether a model can identify suspicious behavior from attack classes that were not included during training.

Table 2.3: Known Attack Detection vs Unseen Attack Detection

Aspect	**Known Attack Detection**	**Unseen Attack Detection**	**Implication for This Study**
Evaluation setting	Closed-set classification.	Open-set, novelty, anomaly, or leave-one-attack-out setting.	The study must evaluate beyond standard closed-set accuracy.
Training data	Includes labeled examples of attack classes to be tested.	Excludes one or more attack classes from training.	The protocol should simulate unseen attack classes.
Model objective	Assign traffic to known classes.	Detect behavior that does not match learned normal or known patterns.	Representation learning and anomaly/novelty logic are needed.
Common metrics	Accuracy, Precision, Recall, F1-score.	Unseen detection rate, AUROC, FPR, FNR, open-set metrics.	Metrics must capture security relevance, not only global accuracy.

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Aspect	Known Attack Detection	Unseen Attack Detection	Implication for This Study
Main risk	High accuracy may reflect memorization of known patterns.	False positives and threshold sensitivity may occur.	Fair comparison and calibration are necessary.

The reviewed literature indicates that strong performance on known attack classification is not sufficient evidence of real-world readiness. Since attackers can modify behavior or use new encrypted communication patterns, a useful IDS should be evaluated under conditions where not all attack classes are available during training [7], [8], [10].

2.12 Open-Set, Novelty, and Anomaly Detection

Open-set detection, novelty detection, and anomaly detection are related but not identical concepts. Open-set detection assumes that the model may encounter unknown classes during testing or deployment. Novelty detection often focuses on identifying samples that do not belong to the learned distribution of known classes. Anomaly detection models normal behavior and detects deviations from it. These concepts are relevant to unseen attack detection because unseen attacks are not fully represented during training.

Several algorithmic strategies can support this problem, including One-Class SVM, Isolation Forest, Local Outlier Factor [Reference needed], autoencoder reconstruction error, threshold-based detection, and distance-based decisions in latent space. Autoencoder-based approaches are particularly common in IDS because reconstruction error can act as an anomaly score when unseen or abnormal traffic is poorly reconstructed [15], [21]. However, anomaly-based methods can also generate false positives, especially when legitimate traffic changes or when the threshold is poorly calibrated [14], [16].

The proposed framework is positioned at the intersection of representation learning and unseen attack detection. Rather than relying only on closed-set classification, it will need a decision strategy that can flag flows or sessions that do not match learned benign or known attack representations. This supports the research hypotheses concerning separability, unseen detection, and false positive control.

2.13 Generalization in IDS

Generalization in IDS refers to the ability of a detection model to maintain meaningful performance beyond the exact training conditions. This may include generalization across attack classes, datasets, traffic protocols, feature extraction tools, network environments, or collection

periods. The term is central to the title of this proposal, but it must be interpreted carefully: the study does not claim universal generalization to all networks; rather, it aims to evaluate generalization under controlled experimental protocols such as leave-one-attack-out or cross-dataset testing.

Dataset-specific evaluation is a major source of overestimated IDS performance. When models are trained and tested on the same dataset distribution, they may learn artifacts rather than attack behavior. Sarhan et al. argue that the lack of standard feature sets across NIDS datasets makes cross-dataset comparison difficult and weakens claims of generalization [5], [6]. Wang et al. also highlight the importance of datasets and comparative evaluation in encrypted malicious traffic detection [4].

Leave-one-attack-out evaluation helps test one aspect of generalization by training without a selected attack class and evaluating whether the model can detect that class as unseen. Cross-dataset evaluation can test whether learned patterns transfer across dataset sources. These evaluation strategies are aligned with the proposed study's aim of developing a generalizable payload-free IDS framework.

2.14 Datasets and Benchmarking for Encrypted Traffic IDS

Datasets are central to IDS research because model performance depends heavily on traffic collection, labeling, feature extraction, class balance, and train-test split policy. IDS datasets may be provided as PCAP files, flow records, NetFlow-like records, or processed feature tables. PCAP files provide raw packet-level evidence but may require significant preprocessing. Flow-level datasets are easier to use for ML evaluation but may hide important preprocessing assumptions.

The reviewed literature identifies several dataset-related challenges: class imbalance, duplicate flows, outdated traffic, unclear split policies, data leakage, dataset-specific artifacts, and inconsistent feature sets [4], [5], [6], [38], [45], [46]. In encrypted traffic IDS, additional challenges include whether the traffic is truly encrypted, whether encryption protocols are diverse, and whether labels distinguish benign traffic, known attack classes, and potential unseen attack categories.

Because leakage between training, validation, and testing partitions can inflate reported IDS performance, dataset preprocessing and split construction must be explicitly documented and checked during methodology design [Reference needed].

For this proposal, dataset selection should be guided by suitability for payload-free flow-level analysis and unseen attack evaluation. Appropriate datasets should contain benign and attack traffic, support flow/session-level representation, provide sufficient labels for known and unseen

evaluation, and allow comparison with baselines. The dataset strategy will be operationalized in Chapter 3, where dataset selection criteria, unit of analysis, preprocessing procedures, train-test split policy, and unseen attack evaluation protocol will be defined in detail.

2.15 Evaluation Metrics for Known and Unseen Attack Detection

Evaluation metrics must reflect the detection objective. In closed-set classification, common metrics include Accuracy, Precision, Recall, F1-score, and confusion matrix. However, Accuracy alone is insufficient in IDS because class imbalance may make a model appear strong while it fails to detect minority attacks or produces excessive false alarms [14], [38].

Security-sensitive metrics include False Positive Rate (FPR), False Negative Rate (FNR), and detection rate. FPR is particularly important because false alarms can overload analysts and reduce operational trust. FNR is equally important because missed attacks can create serious security risk. For unseen attack detection, additional metrics such as Unseen Attack Detection Rate, AUROC, AUPR, open-set F1, or performance drop under leave-one-attack-out evaluation may be required [7], [8], [14].

Generalization evaluation should also report performance across attack classes, datasets, or protocols where possible. The proposed study should therefore use a metrics package rather than a single performance value. This aligns with the Chapter 1 hypothesis that the proposed framework should be evaluated not only for high detection performance but also for false positive control and generalization. The operational definitions and formulas of these metrics are provided in Section 3.22.

2.16 Comparative Literature Matrix

Table 2.4 summarizes the most relevant studies from the reviewed literature. The purpose of the matrix is not to reproduce every detail of each study, but to compare the literature according to the specific dimensions required by this proposal: traffic type, payload-free design, method, dataset, feature type, known/unseen evaluation, metrics, limitations, and relevance.

Table 2.4: Comparative Literature Matrix

Study / Reference	Research Problem	Method / Model	Dataset	Features Used	Known / Unseen Evaluation	Limitations	Relevance to This Study
[1]	AI anomaly detection over encrypted traffic.	SLR of AI methods	30 selected studies	Statistical and traffic features	Mostly reviewed studies	Review only; depends on selected studies.	Direct foundation for encrypted anomaly detection.
[2]	ML-based encrypted traffic analysis and classification.	Survey of ML methods	Multiple studies	Traffic statistics and metadata	Mostly known classes	Classification focus broader than IDS.	Supports background and feature rationale.
[3]	Encrypted traffic analysis applications and techniques.	Survey and taxonomy	Multiple studies	Metadata, characteristics, and related signals	Varied	Survey; not a proposed IDS framework.	Supports taxonomy and context.
[4]	Encrypted malicious traffic detection.	Comparative ML study	Five public sources	Multiple feature sets	Mainly known detection	Dataset and cross-dataset issues remain.	Supports problem and benchmarking gap.
[5]	Standard feature set for NIDS datasets.	NetFlow feature standardization	UNSW, BoT-IoT, ToN-IoT, CSE-CIC	12 and 43 NetFlow features	Known detection	Not specifically unseen encrypted attacks.	Core support for flow-level standardization.
[6]	NetFlow datasets for ML-based NIDS.	Dataset generation	NF-UNSW, NF-BoT, NF-ToN, NF-CSE	NetFlow records	Known detection	Multi-class performance may be lower than original features.	Supports dataset strategy.
[7]	Unknown custom encrypted traffic threats.	Contrastive learning + few-shot fine-tuning	Public + custom data	Payload statistical and learned representations	Few-shot and zero-shot	Uses payload statistics; dataset details require careful reading.	Supports self-supervised unseen detection.
[8]	Encrypted traffic classification with path signatures.	Path signature features + RF	Six public datasets	Packet length sequences	Closed-world and open-world	Traffic classification, not IDS framework.	Supports open-world and sequence representation.
[9]	Feature analysis of encrypted malicious traffic.	ML feature analysis	Encrypted malicious/benign traffic	Traffic features	Known detection	Focus on feature analysis rather than unseen detection.	Supports feature interpretability.
[10]	Mobile encrypted traffic classification.	Deep learning framework	Mobile datasets	Automatically learned traffic features	Includes open-set discussion	Classification focus; not intrusion framework.	Supports open-set motivation.
[11]	Encrypted traffic classification using DL.	SAE + CNN	ISCX VPN-nonVPN	Packet-based features	Known classification	Not payload-free flow-level IDS.	Provides DL baseline context.
[12]	Encrypted classification and intrusion detection.	Deep-Full-Range	Public datasets	Raw traffic features	Known detection/classification	Generalization and unseen detection limited.	Supports DL comparison.
[13]	Anomaly detection in encrypted traffic.	AE, CNN-LSTM, ET-SSL	CIC-Darknet2020, UNSW, QUIC-TLS	Flow-based standardized features	Anomaly-oriented	Publication scope and dataset design need careful analysis.	Highly relevant to proposed representation learning.
[14]	Temporal anomaly detection.	TMG-GRU-VAE + TCI	CIC-IDS-2017/2018	Temporal features	Anomaly detection	Not specifically encrypted-only.	Supports temporal and FPR rationale.
[15]	Hybrid IDS with autoencoder and ML.	AE-GMM + XGBoost/LR	NSL-KDD, UNSW-NB15	Preprocessed features, latent representation	Known detection	Standalone AE weaker for subtle attacks.	Supports hybrid representation approach.
[16]	DL-based IDS systematic review.	Survey of DL IDS	Multiple studies	Multiple feature types	Known and anomaly studies	Review; encrypted traffic not the sole focus.	Supports IDS background and limitations.
[17]	Deep learning for encrypted traffic classification.	Overview	Multiple studies	Statistical and learned features	Known classification	Overview; not unseen IDS-specific.	Foundational context.
[19]	BERT and CNN for encrypted traffic.	BERT + CNN	ISCX-VPN	Packet/byte-level features	Known classification	May use byte-level content; not flow-level IDS.	Supports DL representation discussion.

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Study / Reference	Research Problem	Method / Model	Dataset	Features Used	Known / Unseen Evaluation	Limitations	Relevance to This Study
[20]	Encrypted traffic classification with NIN.	Deep/parallel NIN	ISCX VPN-nonVPN	Header and packet body	Known classification	Packet body use limits payload-free suitability.	Useful comparison but outside core scope.
[21]	Early anomaly traffic detection.	CNN + Autoencoder	Not clearly summarized here	First bytes of early packets	Anomaly detection	First bytes may not be fully payload-free.	Supports early detection limits.
[22]	Two-stage DL NIDS.	LSTM-AE	CICIDS2017, CSE-CICIDS2018	Flow-level features	Known/anomaly-oriented	Encrypted-specific scope limited.	Supports methodology options.
[35]	Feature selection for NIDS.	IGRF-RFE + MLP	UNSW-NB15	Selected tabular features	Known multiclass	Not encrypted/unseen-specific.	Supports feature reduction discussion.
[38]	Big and imbalanced IDS data.	Oversampling + embedding + PCA + ML	UNSW, CIC-IDS-2017/2018	Dimensionality-reduced features	Known detection	Does not focus on unseen encrypted attacks.	Supports imbalance and evaluation caution.

The matrix shows several recurring patterns. First, many encrypted traffic studies focus on classification rather than intrusion detection [8], [10], [11], [17], [19], [20]. Second, many IDS studies focus on known attack detection and standard supervised evaluation [15], [16], [22], [35], [38]. Third, flow-level and NetFlow-like features are strongly supported as practical payload-free representations, but they are not always combined with unseen attack evaluation [5], [6]. Fourth, representation learning appears promising for anomaly detection and behavior modeling, but it still requires careful evaluation under open-set and generalization-aware conditions [7], [13], [14], [15]. These patterns collectively support the research gap developed in Section 2.19.

2.17 Taxonomy of Existing Approaches

The reviewed studies can be grouped into several methodological families. This taxonomy clarifies where the proposed framework is positioned and how it differs from adjacent research traditions.

Table 2.5: Taxonomy of Existing Approaches

Approach Family	Typical Focus	Representative References	Main Limitation Relative to This Study
Payload-based IDS approaches	Inspect packet content, strings, or signatures.	Discussed as traditional approaches in [1], [3].	Not suitable when encrypted payloads are unavailable or privacy-sensitive.
Flow-based / payload-free IDS approaches	Use flow metadata, packet timing, counts, and statistical behavior.	[5], [6], [9], [14].	May still require stronger representation learning and unseen evaluation.

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Approach Family	Typical Focus	Representative References	Main Limitation Relative to This Study
Encrypted traffic classification approaches	Classify applications, services, VPN/non-VPN, Tor/non-Tor, or protocols.	[8], [10], [11], [17], [19], [20].	Classification is not equivalent to unseen attack detection.
ML-based IDS approaches	Use classical ML on tabular traffic features.	[4], [15], [35], [36], [38].	Often closed-set and dependent on labels and dataset-specific features.
DL-based IDS approaches	Learn complex nonlinear patterns using neural networks.	[11], [12], [14], [16], [21], [22], [26].	Can overfit, require data, and lack explainability/generalization evidence.
Representation learning approaches	Learn latent embeddings or compressed behavior representations.	[7], [13], [15], [21].	Need stronger integration with flow-level unseen IDS evaluation.
Open-set / novelty / anomaly approaches	Detect traffic outside known training classes or normal behavior.	[7], [8], [10], [14], [15].	Thresholding and false positives remain challenging.
Generalization / cross-dataset evaluation approaches	Test robustness across datasets or feature standardization.	[4], [5], [6].	Often not directly connected to encrypted unseen attack frameworks.

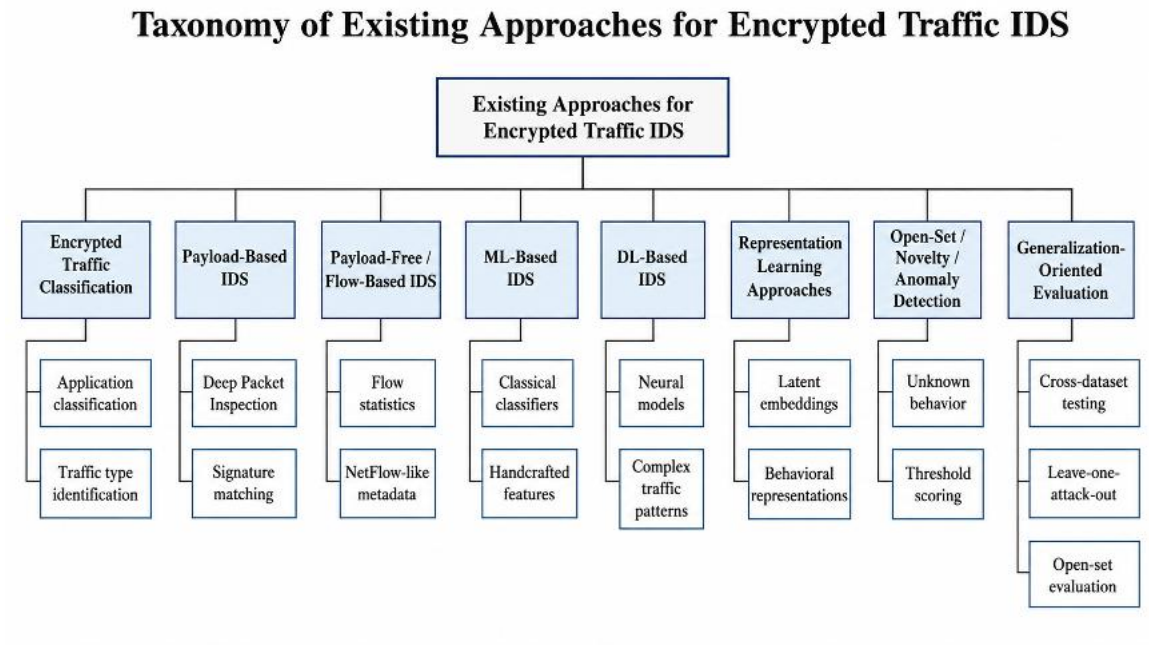


Figure 2.1: Taxonomy of Existing Approaches for Encrypted Traffic IDS

The proposed framework falls primarily into the payload-free flow-based family, but it also integrates representation learning and unseen attack evaluation. This positioning differentiates it from encrypted traffic classification studies that do not directly address intrusion detection, and from conventional IDS studies that focus mainly on known attack classes.

2.18 Critical Synthesis of Literature

Theme 1: Encrypted traffic reduces payload visibility.

The literature consistently shows that encryption limits the effectiveness of DPI and payload-based inspection. This does not mean encrypted traffic cannot be analyzed; rather, the analysis must shift from content to behavior [1], [2], [3]. This theme directly supports the visibility gap identified in Chapter 1.

Implication for this study: this supports the need for payload-free flow-level representation learning under unseen attack evaluation.

Theme 2: Flow-level features provide privacy-preserving signals.

Flow-level features and NetFlow-like metadata provide a practical way to monitor encrypted traffic without decryption. Standardized flow features also improve comparability across datasets [5], [6]. However, flow features may lose payload semantics and may still be affected by dataset-specific artifacts.

Implication for this study: this justifies the use of standardized flow-level and NetFlow-like metadata as the primary evidence source.

Theme 3: ML/DL approaches perform well on known attacks but often assume closed-set evaluation.

ML and DL studies report strong performance in many IDS and encrypted traffic tasks [4], [11], [12], [16], [22]. However, much of this evidence comes from known-class classification. High closed-set accuracy does not necessarily indicate that a model can detect unseen attack classes.

Implication for this study: this supports the need to evaluate models beyond known-class accuracy and closed-set assumptions.

Theme 4: Unseen attack detection requires open-set, novelty, or anomaly-aware design.

Studies addressing open-world classification, contrastive learning, temporal anomaly detection, and autoencoder-based anomaly scoring indicate that unseen attack detection requires different assumptions from standard supervised learning [7], [8], [10], [14], [15]. This suggests that the proposed framework should include an unseen detection strategy rather than only a known attack classifier.

Implication for this study: this motivates an evaluation protocol that explicitly simulates held-out unseen attack classes.

Theme 5: Generalization remains a major limitation.

The literature repeatedly indicates that dataset choice, feature inconsistency, and evaluation protocol can strongly influence reported performance [4], [5], [6]. Therefore, generalization must be evaluated explicitly through leave-one-attack-out, cross-dataset, or controlled open-set pro-

tocols rather than assumed from high accuracy on one dataset.

Implication for this study: this justifies fair baseline comparison, standardized features, and generalization-aware evaluation.

Theme 6: Evaluation protocols and metrics need to be stronger.

Accuracy alone is not sufficient for IDS evaluation because class imbalance and false alarms can distort performance interpretation [14], [38]. A stronger evaluation should include F1-score, Recall, FPR, FNR, AUROC, unseen detection rate, and performance consistency across classes or datasets.

Implication for this study: this supports the use of a broader metric package including FPR, FNR, UADR, and performance drop.

Taken together, these themes show that the proposed study is not positioned as another generic ML or DL IDS model. Instead, it addresses the intersection of four unresolved needs: payload-free encrypted traffic monitoring, flow-level representation learning, unseen attack detection, and generalization-aware evaluation.

2.19 Research Gap Development

The research gap in this proposal emerges from the intersection of multiple limitations rather than from a single missing model. Existing studies have individually addressed encrypted traffic analysis, ML/DL-based IDS, feature engineering, and anomaly detection. However, the reviewed literature suggests that these components are not yet sufficiently integrated into a generalizable payload-free framework for unseen attack detection in encrypted traffic.

Table 2.6: Research Gap Development Matrix

Gap Type	Evidence from Literature	Limitation Identified	How the Proposed Study Addresses It
Visibility Gap	Encrypted traffic reduces payload visibility and limits DPI [1], [2], [3].	Content-based IDS cannot reliably inspect encrypted payloads.	Adopts payload-free detection based on flow-level metadata.
Payload-Free Representation Gap	Flow-level features are practical, but raw features may be limited [5], [6], [9].	Behavioral signals need stronger representation for detection.	Uses flow-level representation learning to model traffic behavior.
Closed-Set Evaluation Gap	Many ML/DL IDS studies evaluate known classes [4], [11], [12], [16].	High accuracy may not imply unseen attack capability.	Uses open-set or leave-one-attack-out evaluation.
Unseen Attack Detection Gap	Open-world and unknown threat studies suggest need for novelty-aware learning [7], [8], [10].	Unseen attacks are not adequately tested in many IDS studies.	Includes an explicit unseen attack detection strategy.

Continued on next page

Gap Type	Evidence from Literature	Limitation Identified	How the Proposed Study Addresses It
Generalization Gap	Dataset-specific features and inconsistent feature sets limit comparability [4], [5], [6].	Models may overfit to dataset artifacts.	Uses standardized flow-level features and generalization-aware evaluation.
Benchmarking and Fair Comparison Gap	Baseline selection and dataset handling influence IDS claims [5], [6], [38].	Comparisons can be unfair if splits, features, or tuning differ.	Defines baseline/SOTA comparison and fair comparison policy in Chapter 3.
Reproducibility Gap	Studies may not fully report preprocessing, seeds, tools, or splits [5], [16].	Results can be difficult to reproduce.	Plans an evidence logging and reproducibility package.

Therefore, the refined research gap is as follows: although previous studies have explored encrypted traffic classification, ML/DL-based intrusion detection, flow-level features, and anomaly detection, the literature still lacks a sufficiently integrated and generalization-aware payload-free IDS framework that uses flow-level representation learning to detect unseen attacks in encrypted network traffic under open-set or leave-one-attack-out evaluation with fair baseline comparison and reproducibility.

2.20 Conceptual Framework of the Proposed Study

The conceptual framework of this study is derived from the research gap developed in the previous section. It connects encrypted network traffic monitoring with payload-free feature extraction, representation learning, known and unseen attack detection, decision logic, and evaluation feedback. The framework is conceptual at this stage; detailed implementation choices will be specified in Chapter 3.

The input component consists of encrypted network traffic represented through flow-level or NetFlow-like metadata. This avoids decryption and supports privacy-preserving detection. The preprocessing and feature extraction component prepares the data through cleaning, encoding, normalization, and extraction of flow-level statistical behavior. The representation learning component transforms the prepared features into a latent space intended to capture behavioral patterns relevant to benign traffic, known attacks, and unseen attacks.

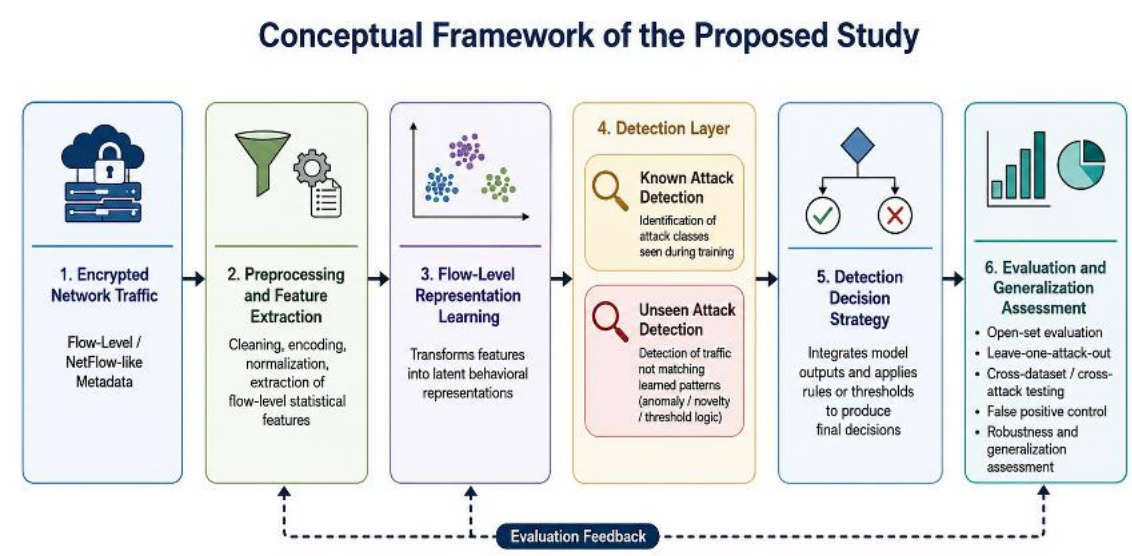


Figure 2.2: Conceptual Framework of the Proposed Study

The known attack detection component addresses supervised or semi-supervised identification of attack classes available during training. The unseen attack detection component addresses traffic that does not match learned known patterns, using anomaly, novelty, threshold, or latent-distance logic. The detection decision strategy integrates model outputs into final decisions, while the evaluation and generalization assessment component measures performance using known-class metrics, unseen attack detection rate, false positive rate, and generalization evidence.

This conceptual framework directly supports the research aim stated in Chapter 1: to design and evaluate a generalizable payload-free intrusion detection framework for unseen attack detection in encrypted network traffic using flow-level representation learning.

2.21 Chapter Summary

This chapter reviewed the literature relevant to encrypted network traffic, intrusion detection systems, payload-free detection, flow-level and NetFlow-based features, ML/DL approaches, representation learning, unseen attack detection, open-set and novelty detection, IDS generalization, datasets, and evaluation metrics. The review showed that encrypted traffic reduces payload visibility and motivates privacy-preserving analysis based on metadata and flow-level behavior. It also showed that while ML and DL approaches can achieve strong results under known-class evaluation, this evidence does not automatically prove unseen attack detection capability.

The comparative literature matrix and taxonomy demonstrated that existing work covers important pieces of the problem, but not always in an integrated manner. Flow-level features support practical payload-free detection, representation learning supports behavioral modeling,

and open-set or anomaly-aware approaches support unseen attack detection. However, the synthesis indicates that a research gap remains in combining these components into a generalizable payload-free IDS framework evaluated under open-set or leave-one-attack-out conditions with fair comparison and reproducibility.

Chapter 3 will translate this literature-based gap into a research methodology and experimental design. It will specify the methodological positioning, framework architecture, dataset selection strategy, preprocessing plan, flow-level feature extraction, representation learning design, known and unseen detection strategies, baseline and SOTA comparison, evaluation metrics, validity threats, ethical considerations, and reproducibility package.

Chapter 3

Research Methodology and Experimental Design

3.1 Introduction

This chapter presents the methodological plan for implementing and evaluating the proposed Generalizable Payload-Free Unseen Detection framework for Encrypted Network Traffic (GPF-UD-ENT). It translates the research problem, research gap, aim, objectives, questions, and hypotheses developed in Chapter 1 and Chapter 2 into an executable experimental design. The central methodological problem is not simply to build a detection model, but to design a defensible evaluation procedure for payload-free unseen attack detection in encrypted network traffic.

The study will adopt a quantitative algorithm/system evaluation approach. It will evaluate the proposed framework using benchmark datasets, controlled preprocessing procedures, flow-level or NetFlow-like features, representation learning, known and unseen attack detection strategies, baseline comparison, open-set or leave-one-attack-out protocols, and numerical performance metrics. This chapter therefore specifies how the framework will be designed, how datasets will be selected and processed, how the experiments will be conducted, how known and unseen attack detection will be evaluated, and how validity, ethics, and reproducibility will be addressed.

3.2 Methodological Positioning

This study will follow a Quantitative Algorithm/System Evaluation Methodology with a Design-Oriented Framework Development Component. The study is quantitative because it will rely on numerical traffic records, measurable variables, controlled experiments, and performance metrics such as precision, recall, F1-score, false positive rate, AUROC, and unseen attack de-

tection rate. It is an algorithm/system evaluation study because the proposed framework will be assessed against baseline models and, where feasible, state-of-the-art approaches under the same experimental conditions.

The design-oriented component appears in the development of the GPF-UD-ENT framework. However, the framework will not be treated as a conceptual contribution alone. It will be evaluated experimentally using benchmark datasets and reproducible protocols. This distinguishes the study from a survey, a purely theoretical framework, or a qualitative study. The methodological positioning is consistent with the literature showing that IDS research commonly requires dataset-based evaluation, baseline comparison, explicit metrics, and controlled experimental protocols [5], [6], [21], [38].

3.3 Research Design

The research design will be a benchmark-based comparative experimental design. The proposed GPF-UD-ENT framework will be evaluated using one or more publicly available intrusion detection datasets that support flow-level analysis, known attack detection, and unseen attack hold-out testing. The design will compare the proposed framework against selected machine learning, deep learning, and anomaly or novelty detection baselines identified in the literature review [4], [15], [16], [21], [22], [26], [27].

The experimental design will include two complementary evaluation settings. The first setting will assess known attack detection, where benign and known attack classes are present during training and testing. The second setting will assess unseen attack detection using open-set or leave-one-attack-out evaluation. In this setting, one or more attack classes will be removed from the training set and introduced only during testing. This design is intended to evaluate whether the framework can detect traffic that does not belong to the learned benign or known attack distributions.

3.4 Study Family Selection

The primary study family is Algorithm/System Evaluation. This is appropriate because the research will design and evaluate an IDS framework using benchmark datasets, baseline models, experimental protocols, and quantitative metrics. The supporting study family is Data-Driven Analytics because the framework will analyze flow-level or NetFlow-like metadata derived from encrypted or encryption-relevant traffic rather than human responses or qualitative interviews.

This design is aligned with the supervisor's methodological guide, which requires the study type, dataset strategy, experimental protocol, baselines, metrics, validity, and reproducibility

procedures to be clearly defined.

The study will not use surveys, interviews, or human-centered measurement instruments. Its unit of evidence will be experimental outputs such as split files, model training logs, confusion matrices, metric tables, error analysis outputs, and reproducibility artifacts. This positioning is aligned with the methodological structure for computing-oriented quantitative research, where ML/IDS studies are evaluated through data, environment, protocols, baselines, and performance analysis.

3.5 Proposed Framework Overview: GPF-UD-ENT

The proposed framework is named GPF-UD-ENT, which stands for Generalizable Payload-Free Unseen Detection for Encrypted Network Traffic. The framework is intended to detect both known and unseen attacks in encrypted network traffic without decrypting payloads or relying on deep packet inspection. It will use payload-free flow-level or NetFlow-like metadata, learn traffic representations, and evaluate detection under known-class and unseen-class conditions.

The design of GPF-UD-ENT is motivated by three findings from the reviewed literature. First, encrypted traffic reduces the visibility of content-based inspection and motivates metadata-based analysis [1], [2], [3]. Second, standardized flow-level features and NetFlow-style datasets can support fairer comparison and more practical extraction than dataset-specific feature sets [5], [6]. Third, representation learning and anomaly-aware detection are useful for modeling complex traffic behavior, although their effectiveness must be evaluated carefully under unseen attack conditions [7], [13], [14], [15].

The framework will include the following components: encrypted traffic input, flow construction, data preprocessing, payload-free flow-level feature extraction, flow-level representation learning, known attack detection, unseen attack detection, detection decision strategy, generalization evaluation, and evidence logging. These components map directly to the conceptual framework developed in Chapter 2.

3.6 Framework Architecture

The architecture of GPF-UD-ENT will be organized into methodological layers rather than implementation-specific code modules. The input layer will receive encrypted traffic traces, flow records, or pre-extracted flow-level datasets. The flow construction layer will convert packet-level traces or PCAP files into network flows or sessions when such raw traces are available. If the selected dataset already provides flow records, this layer will be treated as a verification and harmonization stage rather than a full extraction process.

The payload-free feature layer will extract or select observable metadata such as flow duration, packet counts, byte counts, inter-arrival time, packet length statistics, direction ratios, packet rate, byte rate, and protocol-level fields that do not require payload access. The representation learning layer will transform normalized flow-level features into latent representations designed to capture behavioral patterns beyond raw features. The detection layer will combine a known attack detection strategy with an unseen attack detection strategy. The decision layer will output one of three possible states: benign, known attack, or unseen attack. The evaluation layer will compute closed-set, security-oriented, unseen detection, and generalization metrics.

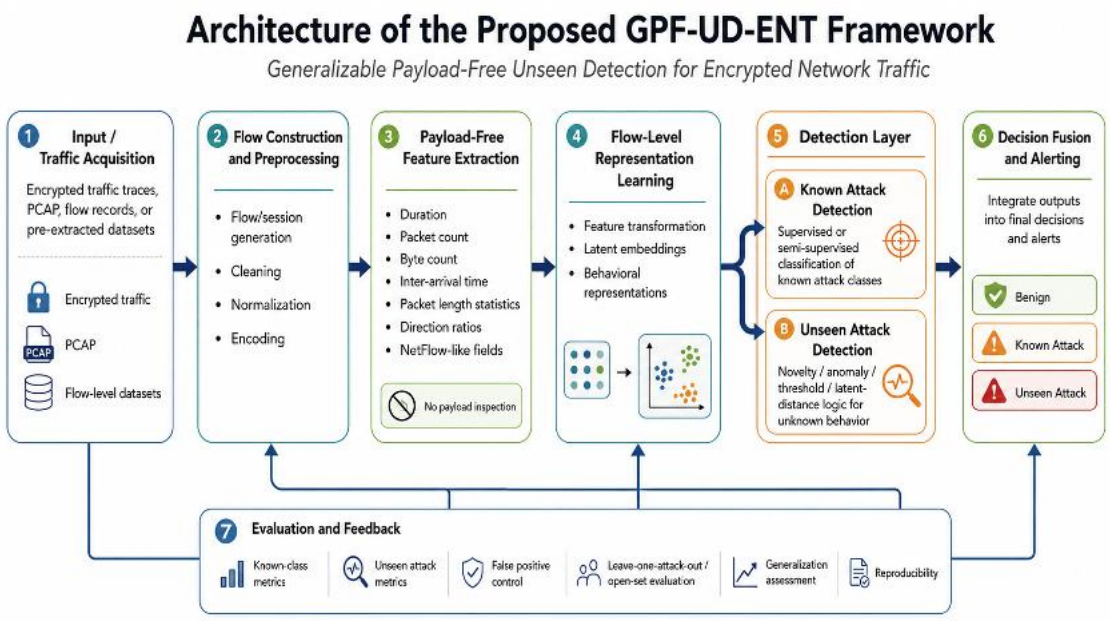


Figure 3.1: Proposed GPF-UD-ENT Framework Architecture

3.7 Dataset Selection Strategy

Dataset selection will follow explicit criteria to ensure that the chosen datasets support the purpose of the study. Because the research focuses on encrypted network traffic and payload-free detection, candidate datasets must either contain encrypted or encryption-relevant traffic, support flow-level analysis, or provide PCAP files from which NetFlow-like metadata can be extracted. The literature shows that dataset quality, feature consistency, labeling, imbalance, and benchmark comparability strongly influence IDS evaluation [4], [5], [6], [38], [45], [46].

Table 3.1: Dataset Selection Criteria

Criterion	Required Condition	Justification
Traffic type	Encrypted, encryption-relevant, or network traffic suitable for flow-level IDS evaluation.	Ensures alignment with the title and research scope.
Encryption relevance	Traffic should include encrypted protocols or scenarios where payload inspection is unavailable or inappropriate.	Supports the payload-free design assumption.
Availability of labels	Benign and attack labels should be available or clearly derivable from dataset documentation.	Required for training, validation, and evaluation.
Benign and attack traffic	The dataset must include normal traffic and multiple attack categories where possible.	Supports both known and unseen attack evaluation.
Attack diversity	The dataset should include several attack classes or families.	Enables leave-one-attack-out evaluation.
Flow-level or PCAP availability	The dataset should provide flow records or PCAP files that can be transformed into flow-level features.	Supports payload-free feature extraction and reproducibility.
Suitability for unseen holdout	Attack categories should be separable enough to hold out one or more classes from training.	Required for open-set or unseen attack testing.
Public availability	The dataset should be publicly accessible or available under research conditions.	Supports reproducibility and supervisor review.
Documentation quality	Class definitions, collection context, features, and preprocessing information should be documented.	Reduces ambiguity and validity threats.
Payload-free compatibility	The dataset should allow evaluation without inspecting or decrypting payload content.	Maintains the privacy-preserving scope of the study.

3.8 Data Sources and Dataset Description

Based on the dataset selection criteria presented in Table 3.1 and the datasets identified in the literature review, this study will consider three primary dataset candidates and one backup dataset for the experimental phase. The primary candidates are NF-CSE-CIC-IDS2018, CIC-IDS2017, and CIC-Darknet2020. NF-CSE-CIC-IDS2018 is prioritized because it provides NetFlow-based features derived from the CSE-CIC-IDS2018 benchmark and is therefore suitable for payload-free flow-level IDS evaluation. CIC-IDS2017 is selected as a secondary benchmark because

it is widely used in IDS research and provides labeled traffic suitable for known and unseen attack evaluation. CIC-Darknet2020 is included as an encryption-relevant candidate because it contains Tor and VPN-related traffic contexts that are relevant to encrypted traffic analysis. As a backup option, UNSW-NB15 or one of its NetFlow-based variants, such as NF-UNSW-NB15, will be considered if the primary datasets do not fully satisfy the required label quality, feature compatibility, or open-set evaluation requirements [5], [6], [13], [14], [21], [38], [45], [46], [47].

Table 3.2: Candidate Datasets for the Proposed Study

Dataset Role	Candidate Dataset	Reason for Selection	Expected Use in This Study	Limitation / Caution
Primary Candidate 1	NF-CSE-CIC-IDS2018	Provides NetFlow-based features derived from a widely used IDS benchmark.	Main candidate for payload-free flow-level IDS evaluation, baseline comparison, and known/unseen attack splits.	Must verify feature schema, class labels, and suitability for leave-one-attack-out evaluation.
Primary Candidate 2	CIC-IDS2017	Widely used IDS benchmark with labeled benign and attack traffic.	Secondary benchmark for reproducible IDS evaluation and comparison with prior studies.	Older dataset; may contain dataset-specific artifacts and requires careful split design.
Primary Candidate 3	CIC-Darknet2020	Contains Tor/VPN-related traffic contexts relevant to encrypted and darknet traffic analysis.	Used to support encryption-relevant traffic analysis and payload-free behavior modeling.	More traffic-classification oriented; label mapping must be reviewed carefully.
Backup Dataset	UNSW-NB15 or NF-UNSW-NB15	Common IDS benchmark; NetFlow variant supports flow-level analysis.	Backup dataset if one of the primary candidates is unavailable or insufficient.	May not be encryption-specific; must be justified as a general IDS benchmark.

The primary candidate datasets will be selected according to label availability, flow-level features, attack diversity, and suitability for LOAO evaluation. No record counts or flow counts will be assumed unless they are explicitly reported in the official dataset documentation.

The final dataset choice will be based on four practical criteria: availability, label quality, compatibility with payload-free flow-level or NetFlow-like feature extraction, and suitability for open-set or leave-one-attack-out evaluation. If the required conditions are satisfied, NF-CSE-CIC-IDS2018 will be used as the main dataset, CIC-IDS2017 will support comparative validation, and CIC-Darknet2020 will support encryption-relevant traffic analysis. The backup dataset will only be used if one of the primary candidates cannot support the experimental protocol.

For each selected dataset, the study will document the dataset name, source, traffic type, encryption context, number of records or flows if reported by the dataset documentation, benign and attack classes, feature availability, labeling method, reason for selection, and limitations. If a dataset provides only packet traces, the flow extraction process will be documented. If a dataset provides precomputed flow-level records, the feature schema and any preprocessing performed by the dataset authors will be reviewed before use. No dataset-specific number of records will be invented; only values reported in official documentation or reviewed literature will be used.

Table 3.3: Dataset Description Template

Dataset Documentation Item	Description to Be Reported	Purpose
Dataset name	To be specified after final selection	The final choice will follow Table 3.1.
Source	Official dataset provider, publication, or repository	Supports traceability and citation.
Traffic type	Encrypted, darknet, IoT, enterprise, or general network traffic depending on selected dataset	Clarifies applicability to encrypted traffic IDS.
Feature availability	Flow-level features, NetFlow-like features, or PCAP for flow extraction	Determines preprocessing and feature extraction plan.
Label structure	Benign and attack classes	Needed for known and unseen attack evaluation.
Selection reason	Alignment with payload-free, flow-level, and unseen attack evaluation	Ensures dataset choice is methodologically justified.
Limitations	Bias, imbalance, outdatedness, missing encryption context, or unclear split policy where applicable	Feeds into threats to validity.

3.9 Unit of Analysis and Unit of Observation

A clear distinction between the unit of observation, unit of analysis, experimental unit, decision unit, and evidence unit is required to avoid methodological confusion. In this study, raw packet-level events may be observed, but the core analysis will be conducted at the flow or session level. The experimental unit will be a complete model run under a defined dataset split and unseen attack holdout condition.

Table 3.4: Unit of Analysis and Unit of Observation

Element	Definition in This Study	Example / Evidence
Unit of Observation	The smallest recorded item, such as a packet, flow record, session event, or extracted feature record.	A packet trace or a single flow feature record.
Unit of Analysis	The entity on which detection decisions and metric calculations are primarily based.	A network flow or encrypted traffic session.

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Element	Definition in This Study	Example / Evidence
Experimental Unit	One complete model run under a fixed split, feature set, method, and attack-holdout condition.	Run 3 with attack class X held out.
Decision Unit	The final output assigned by the framework.	Benign, known attack, or unseen attack.
Evidence Unit	Artifacts used to verify and reproduce the experiment.	Split files, logs, model outputs, confusion matrices, metric tables.

This distinction is important because treating individual packets, flows, sessions, and model runs as equivalent can lead to incorrect conclusions. The primary performance reporting will therefore be based on flow-level or session-level predictions unless a selected dataset requires a different documented unit of analysis.

3.10 Data Preprocessing Plan

The preprocessing plan will be designed to ensure data quality, avoid leakage, and support fair comparison. The study will begin with data inspection to identify feature types, missing values, duplicates, class labels, class imbalance, and inconsistent schema definitions. Duplicate removal will be applied where duplicate flows are confirmed to be artificial or repeated records rather than meaningful repeated events. Missing values will be handled according to feature type and dataset documentation.

Categorical features, if present, will be encoded using a strategy appropriate for the model type. Numerical features will be normalized or standardized, especially before representation learning and distance-based novelty detection. Importantly, transformations that learn from data distributions, such as scaling parameters, will be fitted only on the training data and then applied to validation and test data to reduce data leakage. Where class imbalance is severe, imbalance-handling strategies may be considered, but they will be applied only within the training set to avoid contaminating evaluation results [38].

Scaling, normalization, and imputation parameters will be fitted on the training set only and then applied to validation and testing sets to prevent data leakage [Reference needed].

Train, validation, and test splits will be documented. For unseen attack evaluation, the split policy will also identify which attack class or classes are excluded from training. Feature consistency across datasets will be checked if cross-dataset evaluation is conducted. Any preprocessing differences across datasets will be logged as part of the reproducibility package.

3.11 Flow-Level Feature Extraction

The feature extraction plan will focus on payload-free flow-level and NetFlow-like features. Such features are suitable for encrypted traffic because they describe communication behavior rather than message content. Candidate features include flow duration, packet count, byte count, forward and backward packet counts, forward and backward byte counts, packet length statistics, inter-arrival time, flow rate, byte rate, packet rate, burst behavior, direction-based ratios, and protocol-level metadata that can be extracted without accessing payload content.

The use of flow-level and NetFlow-like metadata is supported by prior work emphasizing the need for standardized features and datasets that improve comparability and practical extraction [5], [6]. The selected features will be reviewed to ensure they do not require payload decryption, deep packet inspection, or application content parsing. When PCAP files are available, the flow extraction process will be documented, including the extraction tool, configuration, timeout rules, and generated feature schema. When precomputed flow-level datasets are used, their feature definitions will be reviewed and aligned with the payload-free scope of this study.

The extracted flow-level features will form the input vector for the representation learning module. In the initial implementation, these features will be used to train an Autoencoder-based representation model as the primary method. A Variational Autoencoder may be considered as an optional extension, while contrastive representation learning may be explored as a secondary option if the dataset structure, label availability, and time constraints allow.

3.12 Representation Learning Design

The representation learning component is central to the proposed framework. Raw flow-level features may contain useful behavioral signals, but they may also be high-dimensional, correlated, noisy, or dataset-dependent. Representation learning will be used to transform normalized flow-level features into latent traffic representations that can support known attack classification and unseen attack detection.

The primary representation learning method selected for the initial implementation will be an Autoencoder-based model. This choice is suitable for the proposed study because Autoencoders can learn compact latent representations from tabular flow-level and NetFlow-like features without requiring payload inspection. The encoder will transform the normalized feature vector into a lower-dimensional latent representation, while the decoder will attempt to reconstruct the original input features. The learned latent representation will then be used as input for known attack detection and unseen attack detection.

A Variational Autoencoder (VAE) will be considered as an optional probabilistic extension of the Autoencoder approach if the selected datasets, feature quality, and implementation time al-

low. The VAE may provide a more structured latent space that can support anomaly-oriented or novelty-aware detection. However, the standard Autoencoder will remain the primary implementation choice because it is simpler, more feasible, and appropriate as an initial representation learning baseline for payload-free flow-level IDS evaluation.

Contrastive representation learning will be treated as a secondary option rather than the main implementation method. This approach may be explored if the selected datasets provide suitable labels or sample-pair construction strategies that allow the model to learn whether two traffic flows are behaviorally similar or different. Contrastive learning may improve latent-space separability, especially between benign traffic, known attacks, and unseen-like behavior, but it requires careful design and will only be included if feasible.

Autoencoder and hybrid representation approaches have been explored in IDS research and can learn compact behavioral patterns from traffic features [13], [15], [21], [22]. Contrastive and self-supervised approaches are relevant because they can reduce reliance on large volumes of labeled malicious encrypted traffic and may improve representation robustness under unknown or few-shot conditions [7], [13]. Therefore, this study will follow a staged representation learning plan: first, raw flow-level features will be used as a baseline; second, an Autoencoder-based latent representation will be developed and evaluated; third, VAE or contrastive learning variants may be examined if feasible.

The quality of the learned representation will be evaluated through downstream known attack detection performance, unseen attack detection performance, latent-space separability where feasible, and ablation comparison against raw flow features. This staged design ensures that representation learning is not treated as a generic concept, but as a specific and testable component of the proposed GPF-UD-ENT framework.

3.13 Known Attack Detection Strategy

The known attack detection strategy will evaluate the ability of the framework to distinguish benign traffic from attack classes that are present during training. Depending on the selected dataset, this component may be formulated as binary classification, multi-class classification, or hierarchical detection. The known detector will be trained using benign samples and known attack classes only. Attack classes designated as unseen in the open-set evaluation will be excluded from the training process. The known attack detection task may be binary or multiclass depending on dataset label granularity.

Baseline classifiers may include Random Forest, Support Vector Machine, Logistic Regression, XGBoost, MLP, CNN-LSTM, or other models already identified in the literature matrix and justified by the selected dataset and feature representation [4], [15], [16], [25], [26], [27], [35], [36], [37]. The known detector may output a predicted class, a probability score, or a confidence

value. These outputs will also inform the final detection decision strategy, particularly when confidence is low or when an input appears outside the known distribution.

3.14 Unseen Attack Detection Strategy

The unseen attack detection strategy will address the core research gap of the study. In this context, an unseen attack is an attack class excluded from training but introduced during testing. The objective is not necessarily to name the exact unseen attack type, but to identify that the traffic behavior does not sufficiently match the learned benign or known attack distributions.

Candidate strategies include open-set thresholding, novelty detection, anomaly detection, distance-based detection in latent space, reconstruction error from an autoencoder, one-class modeling, isolation-based detection, or confidence thresholding. The final strategy will be selected based on the literature review, dataset label structure, and feasibility. If an autoencoder is used, reconstruction error may provide an anomaly score. If metric or contrastive representations are used, distance from known-class clusters or prototypes may support unseen detection. If classifier confidence is used, samples with low confidence may be flagged for novelty assessment.

The unseen detector will be calibrated using validation data without leaking the held-out unseen attack class into training or threshold selection. This is essential because allowing unseen attack samples to influence threshold calibration would weaken the validity of the open-set evaluation.

3.15 Detection Decision Strategy

The detection decision strategy will combine signals from the known detector and unseen detector to produce a final decision. The possible final decisions are benign, known attack, and unseen attack. The decision strategy will be specified before final testing and will not be adjusted using the unseen test class.

Table 3.5: Detection Decision Logic

Input Signal	**Interpretation**	**Decision Role**
Known classifier confidence	High confidence suggests the sample resembles a learned benign or known attack class.	Supports benign or known attack labeling.
Unseen detector score	High novelty or anomaly score suggests the sample does not match the learned distributions.	Supports unseen attack flagging.

Continued on next page

Input Signal	Interpretation	Decision Role
Anomaly / novelty score	Measures deviation from expected traffic behavior.	Used to separate abnormal from normal or known behavior.
Representation distance	Distance from learned latent clusters, prototypes, or known regions.	Supports latent-space open-set decision.
Threshold rule	Predefined validation-based cutoff for novelty or confidence.	Prevents post-hoc tuning on test data.

The thresholds used in the decision strategy will be determined using training and validation partitions only. If multiple thresholds are required, their selection logic will be documented in the reproducibility package. The decision logic will be evaluated through both classification metrics and unseen attack detection metrics.

To improve methodological clarity, Figure 3.2 presents the proposed detection decision strategy that integrates the outputs of the known attack detection module and the unseen attack detection module into a final security decision.

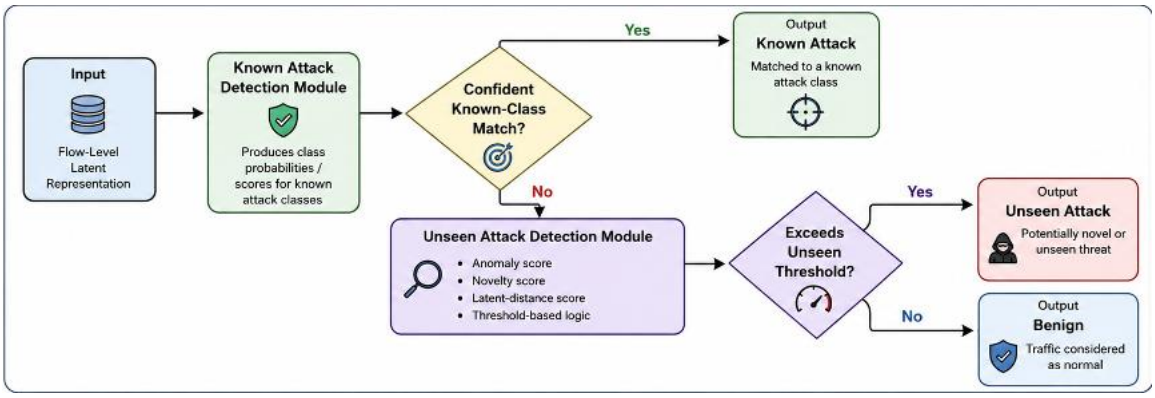


Figure 3.2: Detection Decision Strategy of the Proposed GPF-UD-ENT Framework

As shown in Figure 3.2, the decision strategy first checks whether the known attack detection module produces a confident match to one of the trained attack classes. If a confident match is found, the instance is labeled as a known attack. If not, the unseen attack detection module evaluates the instance using anomaly, novelty, threshold-based, or latent-distance criteria. If the resulting score exceeds the unseen-attack threshold, the instance is labeled as an unseen attack; otherwise, it is classified as benign traffic.

3.16 Generalization Strategy

The term generalizable in this study refers to controlled experimental evaluation beyond a single closed-set setting. It does not imply perfect performance across all networks. The generaliza-

tion strategy will examine how the proposed framework behaves when attack classes, splits, or datasets differ from the training condition. This is important because several IDS studies report strong performance on individual datasets while generalization across datasets, feature sets, or attack distributions remains challenging [4], [5], [6], [38].

Four levels of generalization will be considered. Cross-attack generalization will hold out one attack class and test the ability of the framework to flag it as unseen. Leave-one-attack-out evaluation will repeat this process for each candidate attack class. Cross-dataset generalization will be conducted if the selected datasets have compatible feature schemas and labels. Cross-scenario generalization may be assessed through different splits, traffic groupings, or attack-family holdouts. Performance drop will be reported where feasible to quantify sensitivity to distribution shift.

Cross-dataset evaluation will be conducted only if compatible datasets and feature mappings are available.

3.17 Baseline and SOTA Selection

Baselines and state-of-the-art comparison methods will be selected from the literature matrix and confirmed references. The purpose of baseline selection is not to include every possible IDS model, but to provide a fair and defensible comparison with representative families of approaches: classical ML, deep learning, anomaly or novelty detection, representation learning, and ablation variants of the proposed framework.

To avoid an overly general or unfair SOTA comparison, a method will only be considered a suitable SOTA comparator if it satisfies methodological comparability criteria. Specifically, the selected SOTA method should be evaluated on the same dataset or a closely comparable dataset, address a similar detection task, use compatible feature types such as flow-level, NetFlow-like, or payload-free features, and report evaluation metrics relevant to known and unseen attack detection. When a reviewed study uses a different dataset, a different task, payload-dependent features, or a closed-set-only evaluation setting, it will be treated as related work rather than a direct SOTA comparator.

The selection of direct SOTA comparators will be based on dataset compatibility, task compatibility, feature compatibility, evaluation compatibility, open-set relevance where applicable, and reported metrics. A direct SOTA comparator must therefore use the same or similar dataset type, comparable feature representation, the same or similar IDS task, and compatible evaluation metrics.

Table 3.6: Baseline and SOTA Comparison Plan

Baseline / SOTA Category	**Example Methods**	**Purpose of Comparison**	**Justification from Literature**
Classical ML baselines	Random Forest, SVM, Logistic Regression, XGBoost where suitable.	Assess performance of interpretable and common flow-feature classifiers.	ML classifiers are common in IDS and encrypted traffic studies [4], [9], [35], [36].
Deep learning baselines	MLP, CNN-LSTM, autoencoder-based models where feasible.	Compare against models capable of learning nonlinear patterns.	DL and hybrid IDS models are widely reviewed and evaluated [16], [21], [22], [26].
Anomaly / novelty baselines	One-class, isolation-based, reconstruction-error, or threshold-based detection where feasible.	Provide comparison for unseen attack detection.	Anomaly-oriented approaches are relevant for unseen attack detection [1], [14], [15].
Representation learning baselines	Autoencoder or contrastive representation variants where feasible.	Assess the added value of learned flow representations.	Representation-oriented encrypted traffic work motivates this component [7], [13], [15].
Direct SOTA comparators	Best-performing confirmed studies from the literature matrix that match the dataset, task, features, and evaluation setting.	Provide a stronger comparison against the most relevant prior methods.	A method will be treated as direct SOTA only if it uses the same or comparable dataset, similar IDS/unseen detection task, compatible payload-free or flow-level features, and comparable evaluation metrics.
Ablation baselines	Raw features only; without representation learning; without unseen detector; closed-set only.	Measure the contribution of each component.	Ablation supports fair explanation of framework contribution.
Proposed framework	GPF-UD-ENT full configuration.	Evaluate combined payload-free representation and known/unseen decision strategy.	Addresses the research gap developed in Chapter 2.

This selection policy ensures that the comparison is methodologically fair. A method will not be described as a direct SOTA comparator merely because it reports high performance in the literature. Instead, it must be comparable to the proposed framework in terms of dataset, task, feature representation, and evaluation protocol. This avoids misleading comparisons between payload-based and payload-free methods, traffic classification and intrusion detection tasks, or closed-set and unseen-attack evaluation settings.

3.18 Fair Comparison Policy

A fair comparison policy will be applied to reduce biased evaluation. All models in a comparison group will use the same dataset, same training/validation/testing policy, same feature set where applicable, same preprocessing pipeline, same known/unseen holdout setup, same metrics, same reporting format, and documented random seeds. Hyperparameter tuning will be constrained by the same tuning budget where feasible.

The comparison will avoid using additional data for one method unless the same data is available to all methods in that comparison. Any model-specific requirements will be documented. For example, if a representation learning model uses unlabeled training traffic, the same training partition boundaries will be respected. This policy is necessary because unfair preprocessing, leakage, inconsistent feature sets, or unequal tuning can create misleading performance differences in IDS experiments.

3.19 Experimental Protocol

The experimental protocol will proceed through a sequence of controlled steps. Each step will produce evidence artifacts that can be reviewed later. The protocol is designed to connect dataset selection, payload-free feature extraction, representation learning, known detection, unseen detection, baseline comparison, and generalization assessment.

1. Select and acquire datasets based on the criteria in Table 3.1.
2. Inspect dataset documentation, labels, feature schema, and traffic context.
3. Clean data, remove invalid records, and document missing value handling.
4. Extract or harmonize flow-level and NetFlow-like features.
5. Define train, validation, and test partitions.
6. Define known and unseen attack separation for each experiment.
7. Apply preprocessing and normalization using training data only where appropriate.
8. Train the representation learning component.
9. Train the known attack detector on benign and known attack classes.
10. Calibrate unseen detection thresholds using validation data without unseen test leakage.
11. Train baseline and ablation models under the same comparison policy.

12. Test models on benign, known attack, and unseen attack samples.
13. Compute closed-set, security-oriented, unseen detection, and generalization metrics.
14. Conduct error analysis and performance drop analysis.
15. Log all evidence files, configurations, seeds, and evaluation outputs.

3.20 Unseen Attack Evaluation Protocol

The unseen attack evaluation protocol will use a leave-one-attack-out design where possible. This design is suitable because it simulates a controlled condition in which the model encounters an attack class that was not available during training. The held-out attack class will not be used for model training or threshold calibration. It will appear only in the test set.

1. Select one attack class as the unseen class for the current experiment.
2. Remove all samples of that attack class from the training partition.
3. Train the representation learning component and known detector using benign and known attack classes only.
4. Use validation data to tune model parameters and thresholds without including the held-out unseen attack class.
5. Test the model on benign samples, known attack samples, and the held-out unseen attack class.
6. Record unseen detection metrics, false positives, false negatives, and confusion patterns.
7. Repeat the process for each candidate attack class where dataset labels permit.
8. Report average performance, variation across held-out attacks, and failure cases.

To clarify the evaluation protocol more explicitly, Figure 3.3 presents the step-by-step Leave-One-Attack-Out (LOAO) evaluation procedure used in this study.

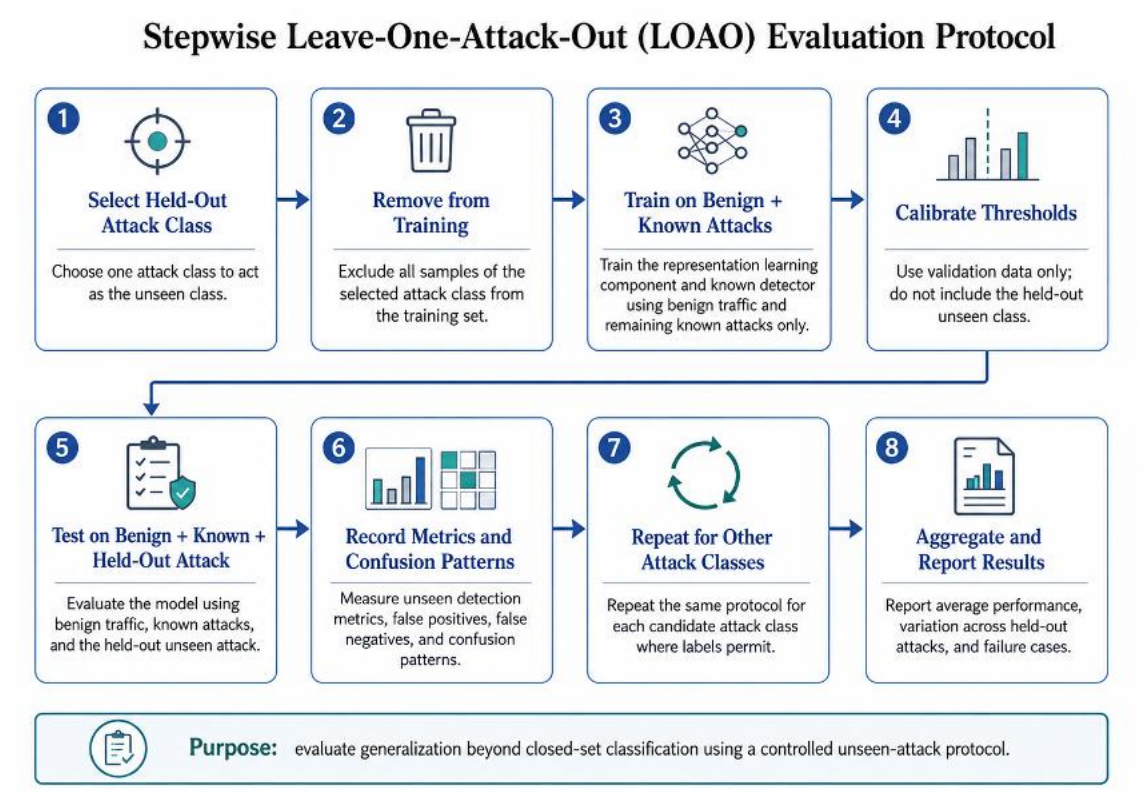


Figure 3.3: Leave-One-Attack-Out Evaluation Protocol

As shown in Figure 3.3, the LOAO protocol begins by selecting one attack class and excluding it completely from training. The framework is then trained using benign traffic and the remaining known attack classes only. During testing, the excluded attack class is reintroduced and treated as an unseen attack together with benign and known attack samples. Performance is then recorded using unseen detection metrics, false positives, false negatives, and confusion patterns. The procedure is repeated across candidate attack classes, and the final results are aggregated to assess generalization and unseen attack detection performance.

3.21 Experimental Variables

Table 3.7: Experimental Variables

Variable Type	Variables	Role in the Study
Independent variables	Feature set; representation learning method; classifier or detector type; dataset; attack-holdout condition; threshold setting.	Define experimental conditions that may influence performance.

Continued on next page

Variable Type	Variables	Role in the Study
Dependent variables	Accuracy; precision; recall; F1-score; false positive rate; false negative rate; AUROC; unseen attack detection rate; performance drop.	Measure detection and generalization outcomes.
Control variables	Preprocessing policy; split policy; random seed; hardware/software environment; evaluation metrics; tuning budget.	Reduce confounding factors and support fair comparison.

The variables will be documented before experiments are executed. Any changes to variable definitions, split policy, or metric selection will be recorded in the evidence log to preserve auditability.

The formulas and operational definitions of the dependent evaluation metrics are specified in Section 3.22 to ensure methodological clarity and reproducibility.

3.22 Evaluation Metrics

The evaluation will use multiple metric categories because accuracy alone is insufficient for IDS evaluation, especially under class imbalance and unseen attack settings. Closed-set metrics will evaluate known attack classification. Security-oriented metrics will quantify false alarms and missed detections. Unseen attack metrics will evaluate whether held-out attack classes can be detected as unfamiliar or suspicious. Generalization metrics will capture performance stability across holdout conditions or datasets where feasible.

For metric calculation, the following notation will be used: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). For unseen attack evaluation, TP_unseen refers to held-out unseen attack samples correctly detected as unseen, while FN_unseen refers to unseen attack samples that are not detected as unseen.

Table 3.8: Evaluation Metrics and Their Purpose

Metric	Formula / Operational Definition	Purpose	Relevance to This Study
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Measures the overall proportion of correctly classified samples.	Useful as a general performance indicator, but insufficient under class imbalance.
Precision	$\frac{TP}{TP+FP}$	Measures the proportion of predicted attack samples that are truly attacks.	Important for reducing false alarms and improving alert reliability.
Recall / Detection Rate	$\frac{TP}{TP+FN}$	Measures the proportion of actual attack samples correctly detected.	Critical for evaluating security monitoring effectiveness.

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Metric	Formula / Operational Definition	Purpose	Relevance to This Study
F1-score	$\frac{2\times Precision\times Recall}{Precision+Recall}$	Measures the harmonic mean of Precision and Recall.	Balances detection capability and false alarm concerns.
Confusion Matrix	A matrix showing TP, TN, FP, and FN across classes.	Provides detailed counts of correct and incorrect decisions.	Supports error analysis, confusion pattern analysis, and failure case interpretation.
False Positive Rate (FPR)	$\frac{FP}{FP+TN}$	Measures the proportion of benign samples incorrectly flagged as attacks.	Important for assessing alert fatigue and operational feasibility.
False Negative Rate (FNR)	$\frac{FN}{FN+TP}$	Measures the proportion of attack samples missed by the detector.	Important for assessing security risk caused by undetected attacks.
Unseen Attack Detection Rate (UADR)	$\frac{TP_{unseen}}{TP_{unseen}+FN_{unseen}}$	Measures the proportion of held-out unseen attack samples correctly detected as unseen or anomalous.	Directly evaluates the core research objective of unseen attack detection.
AUROC	Area under the Receiver Operating Characteristic curve.	Measures ranking ability across different thresholds.	Useful for threshold-sensitive known or unseen detection where applicable.
AUPR	Area under the Precision-Recall curve.	Measures precision-recall performance under class imbalance.	Useful when attack or unseen classes are rare.
Performance Drop	Closed-set performance − unseen/open-set/cross-dataset performance.	Measures the reduction in performance under more difficult evaluation settings.	Evaluates generalization robustness beyond closed-set testing.
Training / Inference Time	Time required for model training and prediction.	Measures computational cost where feasible.	Provides insight into operational feasibility.

For multi-class known attack evaluation, macro-averaged or weighted Precision, Recall, and F1-score may be reported where appropriate. For unseen attack evaluation, UADR will be treated as a core metric because it directly measures whether held-out attack classes are detected as unseen or anomalous. These formulas will be applied consistently across baseline models, ablation variants, and the proposed GPF-UD-ENT framework.

3.23 Statistical and Experimental Analysis Plan

The analysis plan will combine descriptive, comparative, and error-focused analysis. Descriptive statistics will summarize dataset composition, feature distributions, class imbalance, and experimental outputs. Model performance will be reported using mean and standard deviation across repeated runs where feasible. Confidence intervals may be reported if the design and number of runs support them.

Comparative analysis will present the proposed framework against baselines and ablation variants. Confusion matrices, ROC curves, precision-recall curves, and metric tables will be used where applicable. Ablation analysis will assess the contribution of representation learning, unseen detection, and decision logic. Error analysis will examine false positives, false negatives, and attack classes that are difficult to detect. Performance drop analysis will be used to assess generalization across holdout conditions or datasets. Statistical tests may be used only when the repeated-run structure and assumptions make them appropriate; otherwise, the study will rely on transparent metric reporting and repeated-run comparisons.

Statistical tests will be considered when repeated experimental runs provide sufficient observations for meaningful comparison.

3.24 Experimental Environment

The experimental environment will be documented to support reproducibility. The final environment details will be specified during implementation and will include operating system, CPU, GPU if used, RAM, Python version, major libraries, dataset versions, random seeds, feature extraction tools, and execution scripts. Candidate libraries may include pandas, NumPy, scikit-learn, PyTorch or TensorFlow, and visualization libraries, depending on the final method selection. These tools will not be treated as methodology by themselves; they will support the experimental protocol described in this chapter.

Table 3.9: Experimental Environment Documentation Plan

Environment Element	**Planned Documentation**	**Purpose**
Operating system	To be specified during implementation	Required for reproducibility.
Hardware	CPU/GPU/RAM to be specified	Influences training and inference time.
Programming language	Python version to be specified	Common environment for ML/IDS experiments.
Libraries	pandas, NumPy, scikit-learn, PyTorch/TensorFlow if used	Supports preprocessing, modeling, and evaluation.
Flow extraction tool	To be specified if PCAP-to-flow extraction is performed	Affects feature definitions and reproducibility.
Random seeds	Fixed and documented seeds where applicable	Supports repeatability.
Dataset version	Version or download date to be recorded	Prevents ambiguity in benchmark use.

3.25 Evidence Logging and Reproducibility Package

A reproducibility package will be prepared during implementation. It will not necessarily publish restricted datasets, but it will preserve all artifacts needed to audit the experimental process. The evidence package will include dataset source notes, preprocessing scripts, feature extraction scripts, feature schema files, train/validation/test split files, leave-one-attack-out configurations, model parameters, hyperparameter settings, training logs, evaluation outputs, confusion matrices, figures, error analysis files, environment configuration, random seeds, and notes on

excluded records.

Evidence logging is important because IDS studies are sensitive to split policy, preprocessing, feature schema, imbalance handling, and threshold calibration. Without transparent logging, reported results may be difficult to reproduce or compare fairly. The reproducibility package will therefore function as an audit trail connecting the research problem, experimental design, model outputs, and reported metrics.

3.26 Threats to Validity

Table 3.10: Threats to Validity and Mitigation Strategies

Validity Threat	**Description**	**Mitigation Strategy**
Internal validity	Data leakage, inconsistent preprocessing, threshold tuning on test data, or unfair baseline configuration.	Fit preprocessing only on training data; separate validation from test; use documented split policies; apply fair comparison rules.
Construct validity	Metrics or features may not fully represent real intrusion detection effectiveness.	Use multiple metric categories; include FPR, FNR, unseen detection rate, and performance drop; document feature meaning.
External validity	Public datasets may not represent all real encrypted enterprise traffic or modern attack behavior.	Discuss dataset scope; use multiple datasets if feasible; avoid claims of universal generalization.
Conclusion validity	Small number of runs, class imbalance, or unstable thresholds may lead to overinterpretation.	Use repeated runs where feasible; report variation; perform error analysis and cautious interpretation.
Reproducibility threats	Tool versions, feature extraction differences, dataset version changes, or undocumented random seeds.	Log environment, scripts, seeds, dataset versions, and feature extraction settings.

3.27 Ethical and Privacy Considerations

The study will be conducted for defensive cybersecurity research. It will not involve unauthorized traffic collection, payload decryption, malware deployment, or offensive exploitation. The proposed framework is payload-free and will rely on flow-level metadata rather than reading en-

crypted content. This reduces privacy risk and aligns with the motivation for privacy-preserving encrypted traffic monitoring.

Datasets will be used according to their licenses and research terms. If public datasets are used, the study will cite the relevant dataset publications and documentation. If any dataset contains potentially sensitive metadata, the study will use it only for research purposes and will avoid reporting information that could identify individuals, organizations, or private communications. The thesis will not provide harmful operational instructions for attacking networks.

3.28 Feasibility and Risk Management

Table 3.11: Feasibility and Risk Management Plan

Risk	**Potential Impact**	**Mitigation Plan**
Dataset not suitable	May prevent alignment with encrypted or payload-free scope.	Use explicit selection criteria and maintain backup datasets from the literature.
Insufficient unseen attack classes	May limit leave-one-attack-out evaluation.	Use attack family holdout or cross-scenario holdout where labels permit.
Class imbalance	May bias metrics and reduce minority attack detection.	Use stratified splits where feasible; report FPR, FNR, recall, and AUPR when appropriate.
Model overfitting	May inflate results on a single dataset.	Use validation data, ablation, repeated runs, and generalization assessment.
High false positive rate	May reduce operational relevance.	Calibrate thresholds using validation data and report FPR explicitly.
Weak cross-dataset generalization	May limit the generalizable claim.	Report performance drop and avoid overclaiming beyond tested conditions.
Computational limitations	May constrain DL or contrastive learning experiments.	Start with feasible models and scale complexity only when justified.
Limited time	May affect number of datasets or baselines.	Prioritize core protocol: payload-free features, representation learning, unseen holdout, and key baselines.

3.29 Expected Outcomes

The study is expected to produce a proposed GPF-UD-ENT framework architecture, a benchmark-based experimental evaluation pipeline, a payload-free flow-level feature representation strategy, unseen attack detection results, baseline comparison results, generalization analysis, and validity and reproducibility documentation. These outputs are expected to support the proposed contribution without claiming guaranteed superior performance in all scenarios.

The expected scientific outcome is a defensible methodology for evaluating unseen attack detection in encrypted traffic using payload-free flow-level representations. The expected practical outcome is a privacy-preserving IDS evaluation framework that can be studied without payload decryption. The expected methodological outcome is a reproducible experimental process that documents dataset selection, feature extraction, open-set protocol, baselines, metrics, threats to validity, and evidence logging.

3.30 Proposed Timeline

Table 3.12: Proposed Research Timeline

Period	Main Task	Expected Output
Weeks 1-2	Finalize RQs, hypotheses, methodology, and dataset criteria.	Approved methodology plan.
Weeks 3-5	Complete targeted literature refinement and finalize dataset candidates.	Updated literature matrix and dataset shortlist.
Weeks 6-8	Acquire datasets and inspect labels, schema, and feature availability.	Dataset documentation report.
Weeks 9-11	Extract or harmonize flow-level features and implement preprocessing.	Cleaned feature datasets and preprocessing scripts.
Weeks 12-14	Design and implement representation learning component.	Initial latent representation model.
Weeks 15-16	Implement known and unseen detection strategies.	Detector modules and threshold strategy.
Weeks 17-18	Train baselines and ablation models.	Baseline results and ablation setup.
Weeks 19-20	Conduct leave-one-attack-out and generalization experiments.	Unseen detection and generalization results.

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Period	Main Task	Expected Output
Weeks 21-22	Analyze metrics, errors, validity threats, and performance drops.	Results tables, figures, and error analysis.
Weeks 23-24	Revise methodology, finalize evidence package, and prepare thesis/proposal text.	Supervisor-ready methodology and experimental package.

3.31 Chapter Summary

This chapter translated the research gap into an executable methodology. It positioned the study as a quantitative algorithm/system evaluation with a design-oriented framework development component. It described the proposed GPF-UD-ENT framework, its architecture, dataset selection strategy, unit of analysis, preprocessing plan, flow-level feature extraction, representation learning design, known and unseen attack detection strategies, decision logic, generalization strategy, baselines, fair comparison policy, experimental protocol, unseen attack evaluation protocol, variables, metrics, analysis plan, environment, reproducibility package, validity threats, ethical considerations, feasibility plan, expected outcomes, and timeline.

The methodology is intended to answer the research questions from Chapter 1 and operationalize the research gap developed in Chapter 2. The next stage of the research will implement and evaluate the proposed framework using the protocol described in this chapter.

Conclusion

This proposal has developed a focused research direction for designing and evaluating a generalizable payload-free intrusion detection framework for unseen attack detection in encrypted network traffic using flow-level representation learning. The proposed study is positioned within cybersecurity, network intrusion detection, encrypted traffic analysis, and machine learning-based evaluation. Its central contribution is not a claim of universal detection, but a structured and reproducible framework for evaluating whether payload-free flow-level representations can support known and unseen attack detection under open-set or leave-one-attack-out conditions.

The proposal is grounded in a research problem identified across the literature: encrypted traffic limits payload visibility; many IDS approaches remain closed-set; dataset-specific features weaken generalization; and unseen attack detection requires stronger evaluation beyond standard accuracy. The proposed methodology addresses this gap through explicit dataset selection criteria, flow-level feature extraction, representation learning, known and unseen detection strategies, fair baseline comparison, evaluation metrics, validity analysis, ethical safeguards, and reproducibility planning.

The next phase of the research will finalize dataset selection, implement the proposed experimental protocol, compare the GPF-UD-ENT framework with selected baselines, and analyze the ability of the framework to detect unseen attack behavior without payload inspection or decryption.

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Appendix A

Dataset Evaluation Checklist

Table A.1: Dataset Evaluation Checklist

Checklist Item	**Status**
Dataset name recorded	Yes / No
Official source documented	Yes / No
Traffic type described	Yes / No
Encryption relevance verified	Yes / No
Benign labels available	Yes / No
Attack labels available	Yes / No
Attack classes suitable for holdout	Yes / No
Flow-level features available or extractable	Yes / No
Payload-free analysis possible	Yes / No
Dataset limitations documented	Yes / No

Appendix B

Leave-One-Attack-Out Experiment Sheet

Table B.1: Leave-One-Attack-Out Experiment Sheet

Experiment Element	**Planned Record**
Held-out attack class	To be specified
Training classes	Benign + known attack classes only
Validation classes	No held-out unseen attack leakage
Testing classes	Benign + known attacks + held-out unseen attack
Threshold calibration policy	Validation data only
Metrics to record	F1, FPR, FNR, AUROC, unseen attack detection rate, performance drop
Evidence files	Split files, model logs, metric tables, confusion matrix

Appendix C

Reproducibility Evidence Checklist

Table C.1: Reproducibility Evidence Checklist

Evidence Artifact	**Planned Status**
Dataset source notes	To be saved
Preprocessing scripts	To be saved
Feature extraction configuration	To be saved
Train/validation/test split files	To be saved
Attack holdout configuration	To be saved
Model parameter settings	To be saved
Hyperparameter search notes	To be saved
Random seeds	To be saved
Environment configuration	To be saved
Evaluation outputs	To be saved
Error analysis files	To be saved

Appendix D

Risk Register Template

Table D.1: Risk Register Template

Risk	Level	Planned Mitigation
Dataset incompatibility	Medium	Maintain alternative datasets and verify feature schema early.
Insufficient unseen attack classes	Medium	Use attack-family holdout when individual class holdout is not feasible.
Class imbalance	High	Report imbalance-aware metrics and apply training-only balancing if justified.
High false positives	High	Use validation-based threshold calibration and report FPR explicitly.
Limited compute resources	Medium	Prioritize feasible models and ablations.
Weak generalization	Medium	Report performance drop transparently and avoid overclaiming.